TUTORIAL FOR THE PRACTICAL ACTIVITY ON POTENTIOMETRIC SURFACE MAPPING

Course: 4145/5145 - Groundwater Resources Engineering (Fall, 2026)

1. Objective

The main objective of this practical activity is to learn how to create and interpret a groundwater level map using data from monitoring wells.

By the end of this activity, you should be able to:

- Prepare and organize groundwater-well data for spatial analysis.
- Calculate the water table elevation (hydraulic head) at each well.
- Create an equipotential contour map representing the water table (potentiometric surface).
- Use the equipotential lines to identify and interpret the direction of groundwater flow.

Remember: under typical conditions, groundwater flows from areas of higher hydraulic head toward areas of lower hydraulic head, approximately perpendicular to the equipotential lines.

2. Study Area

For this activity, we will work with the Catawba River Basin in North Carolina, USA. All students will use the same predefined study area boundary so that we can select a consistent set of groundwater monitoring wells and directly compare the resulting maps.

3. Groundwater Data Acquisition from the U.S. Geological Survey (USGS)

The groundwater monitoring data used in this activity are originally provided by the U.S. Geological Survey (USGS) and will be accessed through the Global Groundwater Information System (GGIS). GGIS is an online groundwater information platform managed by the International Groundwater Resources Assessment Centre (IGRAC). It provides access to groundwater monitoring information from multiple national and international sources.

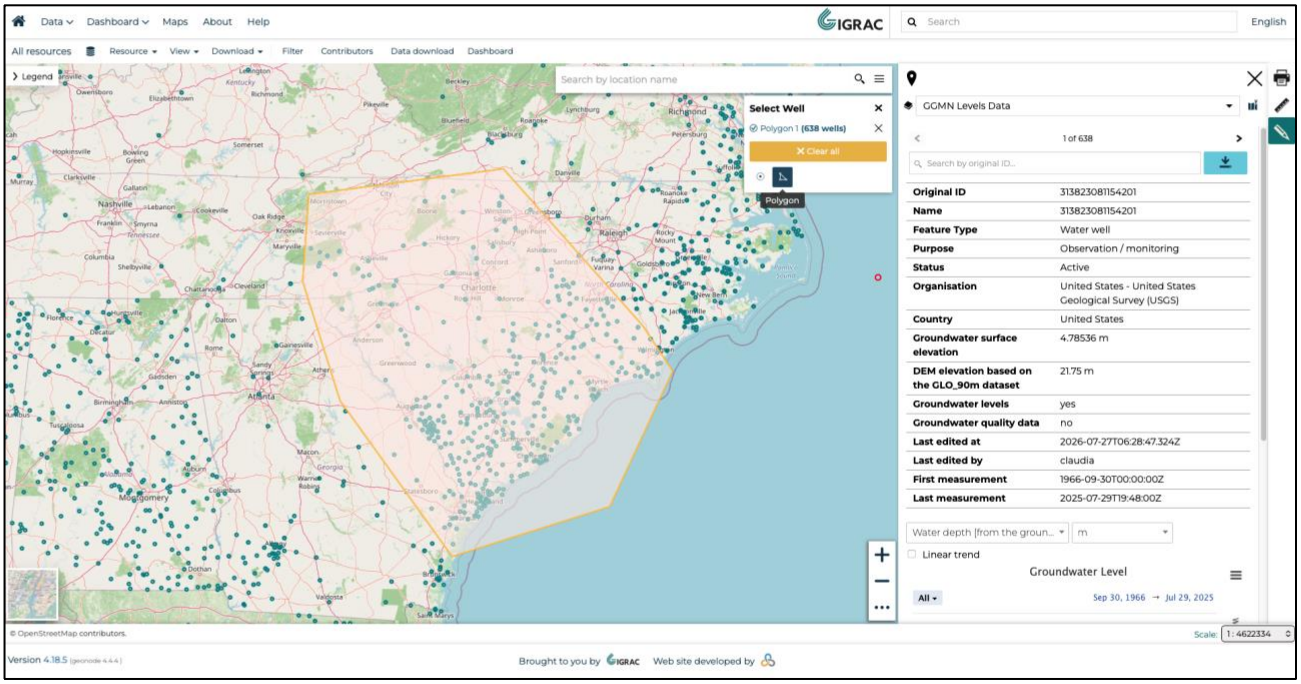
3.1. Extrating from IGRAC

In the following steps, you will learn how to select, download, filter, and prepare the groundwater-well data needed to create the water-table map in ArcGIS.

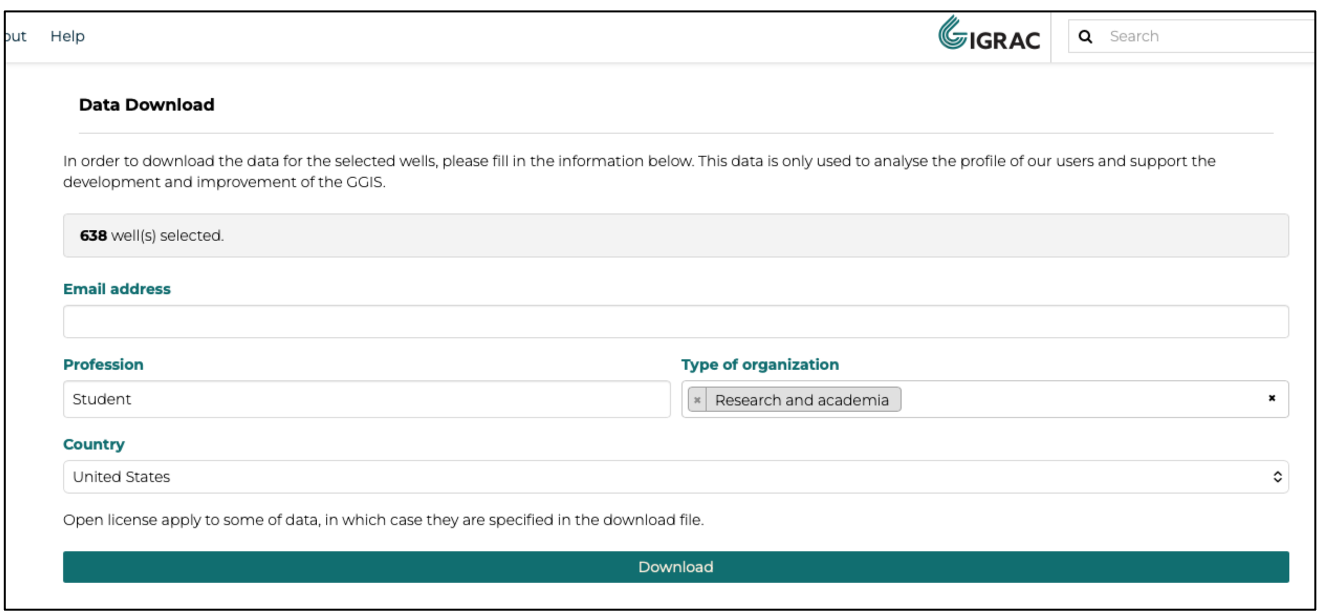
> **Direct Access URL:** https://ggis.un-igrac.org/catalogue/#/map/2504

Step-by-step extraction from GGIS:

1. Access the interactive web map interface. The map is displayed in the center, with the data visualization menu located on the right panel.
2. In the right menu, click on “Select Well”, and “Polygon”. Then, select the area that you want to analyze. In our case, we will click around the area of the Catawba River Basin. Double click to finish the polygon, and the right panel will show as follows. Note: The selection tool might operate by country or for each well, if needed.



3. Click on the download button in the right panel.
4. The download page will ask you for your information, such as e-mail address, profession, type of organization and country you are from. After filling your data, click on “Download”.



5. After generating the files, you may download the .zip archive, which contains timeseries from the wells, and information such as coordinates, elevation, the data origin, as well as is the aquifer is confined or unconfined. This information will be important for us in the next steps.

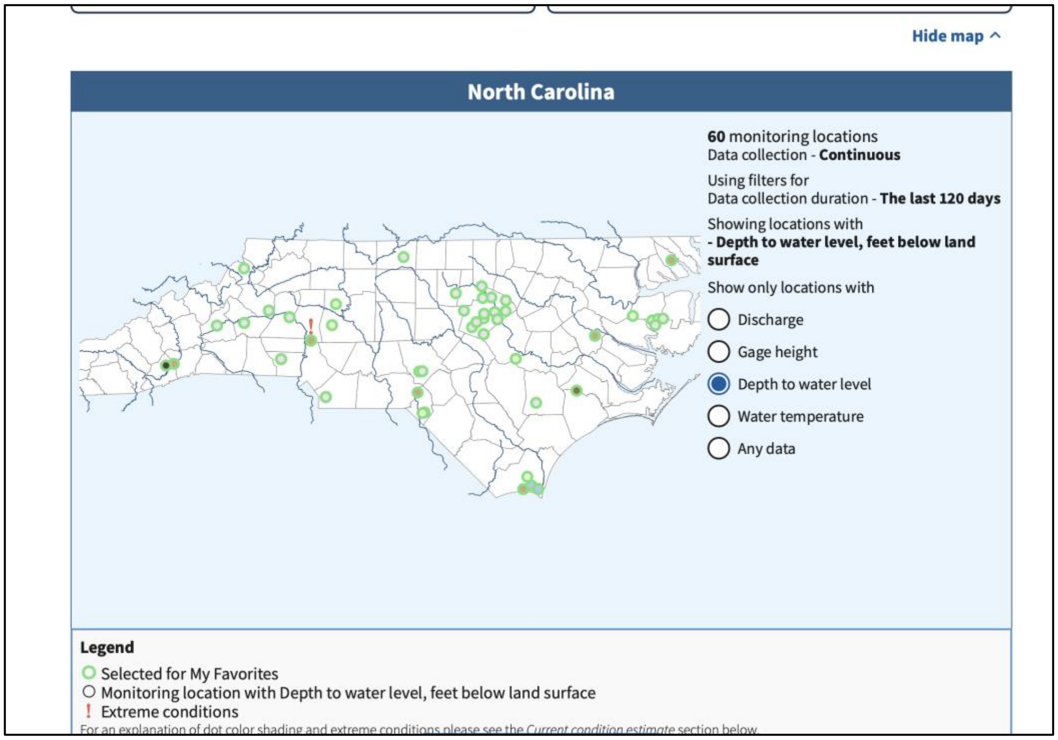
Directory Setup Recommendation: Extract all files from the .zip into a local project folder. Ensure that the directory path and folder names do not contain special characters, accents, or blank spaces (e.g., use C:/hidrology/practicalactivity).

Supplementary Shapefiles (available in Canvas):

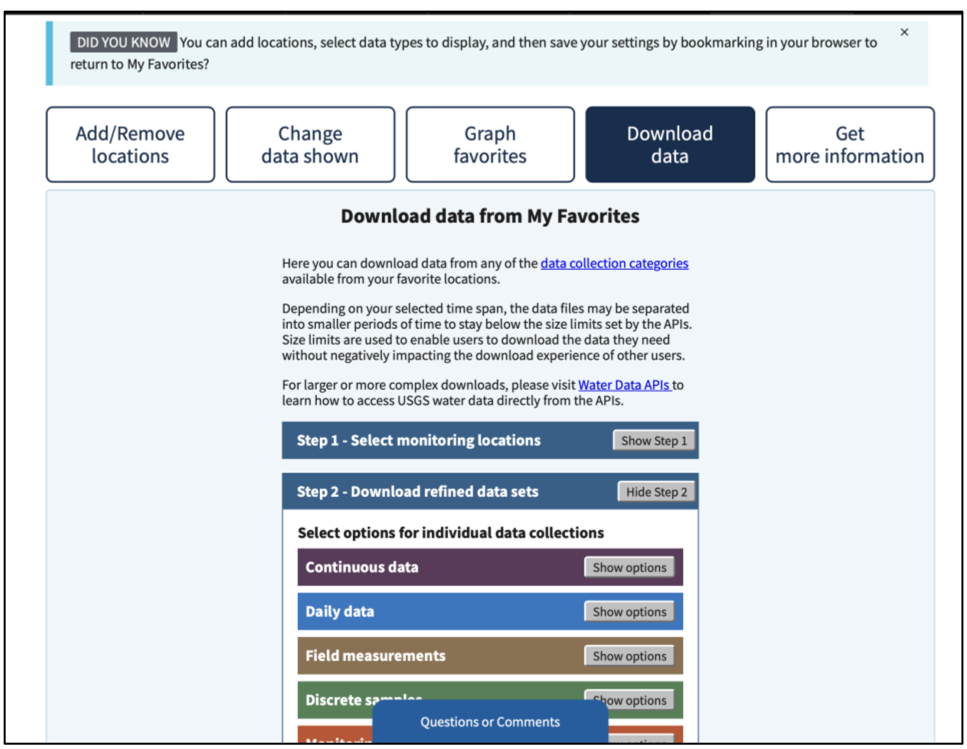
- catawba_river_basin.rar: Contains the shapefile boundary delineating the CRB (boundaries_catawba.shp).

3.2. Alternative: directly from USGS

Alternatively, we can download the data from the USGS. Here, you can access the link for North Carolina: https://waterdata.usgs.gov/state/North%20Carolina/#dataTypes=72019&groupBy=huc8. On the website, you might choose the monitoring locations for water depth and other types of data, and choose each one by county or even basin for your favorite selection.



After choosing the locations, go to My Favorites page and Download data, following the steps.



4. Processing Well Data

One important point is that we need only one representative water-depth value for each well to create the Potentiometric / Water Table Map. However, groundwater levels change over time, so instead of selecting a single measurement from a particular day, we calculate the median water depth for each well over a defined time period. For this activity, we use measurements from October 1, 2024, through September 30, 2025, which represents the most recent complete hydrological year. Using the median reduces the influence of unusually high or low groundwater-level measurements and provides a representative value for each well during the study period.

A Python script was used to calculate the median water depth for each well. The script and supporting files are available in the Google Drive folder and the GitHub repository. Both are available on Canvas.

Then, we have a .csv with well ID, latitude, longitude, elevation and median water depth. We must filter some of the wells now.

4.1. Filtering wells

As expected, not all wells contain the information we need for this activity. Some wells may be missing land-surface elevation or water depth measurements during the selected study period. Furthermore, some wells are placed in confined aquifers. Because our goal is to develop a water table map for an unconfined aquifer, these confined wells should not be included in the analysis. Therefore, before creating the map, we need to filter and clean the dataset so that we keep only wells with the necessary information and the appropriate aquifer type.

4.1.1. Remove wells with missing data

Open the dataset in Excel and use the Filter tool to identify and remove rows with empty cells in the variables required for the analysis, particularly land-surface elevation and water depth.

After this step, every remaining well should have the information necessary to calculate its water-table elevation (hydraulic head).

ID	lat	long	elevation	median_water_depth
3.16E+14	31.9852145	-81.0198337	2.892552	2.56032
3.16E+14	31.99742964	-81.269834	5.428488	1.045464
3.20E+14	32.0341019	-80.9031665	-0.21336	1.746504
3.35E+14	33.8934575	-82.36262688	121.92	11.454384
3.42E+14	34.41111111	-80.74416667	91.397328	16.120872
3.42E+14	34.41113056	-80.74441111	91.397328	14.087856
3.43E+14	34.5181111	-80.2894167	92.6592	4.0386
3.44E+14	34.62066938	-82.4820722	239.268	0.917448
3.44E+14	34.67430433	-81.41509653	202.692	28.038552
3.44E+14	34.72597222	-80.84333333	155.7528	7.97052
3.44E+14	34.726	-80.84333333	156.0576	8.761476
3.45E+14	34.84759273	-83.07154042	329.184	9.081516
3.45E+14	34.86262792	-81.84121266	188.707776	13.2969
3.46E+14	34.97055556	-79.52833333	131.15544	9.244584
3.46E+14	34.97516667	-81.05875	206.98968	6.556248
3.46E+14	34.97519444	-81.05880556	206.9592	5.611368
3.52E+14	35.28583333	-82.72805556	663.48864	4.255008

4.1.1. Filtering unconfined wells

In the files downloaded from G-GIS, you will find a file named well.ods, which contains additional information about the wells. Open the well.ods file and locate the "Hydrogeology" tab. Copy this entire tab into your new filtered Excel file as a new worksheet. We will use this information to determine if each well is associated with a confined or unconfined aquifer. To match the wells in your filtered dataset with the information in the Hydrogeology worksheet, we can use the =VLOOKUP() function. Use the Well ID as the common identifier between the two worksheets.

Important: The Well ID columns in the two worksheets must have the same data format for VLOOKUP() to work correctly. For example, if the IDs are stored as numbers in one worksheet but as text in the other, Excel may not recognize them as matching. If necessary, convert the Well ID cells to Number format. Select all the cells in the Well ID column. If Excel displays a warning icon next to the selected cells, click it and choose "Convert to Number."

Once the Well IDs have the same format in both worksheets, use VLOOKUP() to retrieve the hydrogeologic/aquifer information for each well. You can then identify and remove the wells associated with confined aquifers, keeping only the wells appropriate for our water-table mapping activity.

ID	lat	long	elevation	median_water_depth	Confinement
3.16E+14	31.9852145	-81.0198337	2.892552	2.56032	Unconfined
3.16E+14	31.99742964	-81.269834	5.428488	1.045464	Unconfined
3.20E+14	32.0341019	-80.9031665	-0.21336	1.746504	Unconfined
3.35E+14	33.8934575	-82.36262688	121.92	11.454384	Unconfined
3.42E+14	34.41111111	-80.74416667	91.397328	16.120872	Unconfined
3.42E+14	34.41113056	-80.74441111	91.397328	14.087856	Unconfined
3.43E+14	34.5181111	-80.2894167	92.6592	4.0386	Unconfined
3.44E+14	34.62066938	-82.4820722	239.268	0.917448	Unconfined
3.44E+14	34.67430433	-81.41509653	202.692	28.038552	Unconfined
3.44E+14	34.72597222	-80.84333333	155.7528	7.97052	Unconfined
3.44E+14	34.726	-80.84333333	156.0576	8.761476	Unconfined
3.45E+14	34.84759273	-83.07154042	329.184	9.081516	Unconfined
3.45E+14	34.86262792	-81.84121266	188.707776	13.2969	Unconfined
3.46E+14	34.97055556	-79.52833333	131.15544	9.244584	Unconfined
3.46E+14	34.97516667	-81.05875	206.98968	6.556248	Unconfined
3.46E+14	34.97519444	-81.05880556	206.9592	5.611368	Unconfined
3.52E+14	35.28583333	-82.72805556	663.48864	4.255008	Unconfined

Once you have identified the confinement status of each well, copy the entire column and use **Paste Special → Values** to paste only the results. This removes the VLOOKUP() formulas while keeping the confinement information in your dataset.

After pasting the values, you can delete the additional “Hydrogeology” worksheet, since it is no longer needed. This will keep your file simpler and easier to work with.

Finally, filter the confinement status column and remove all wells classified as confined. Your final dataset should contain only unconfined wells with the required elevation and water-depth information.

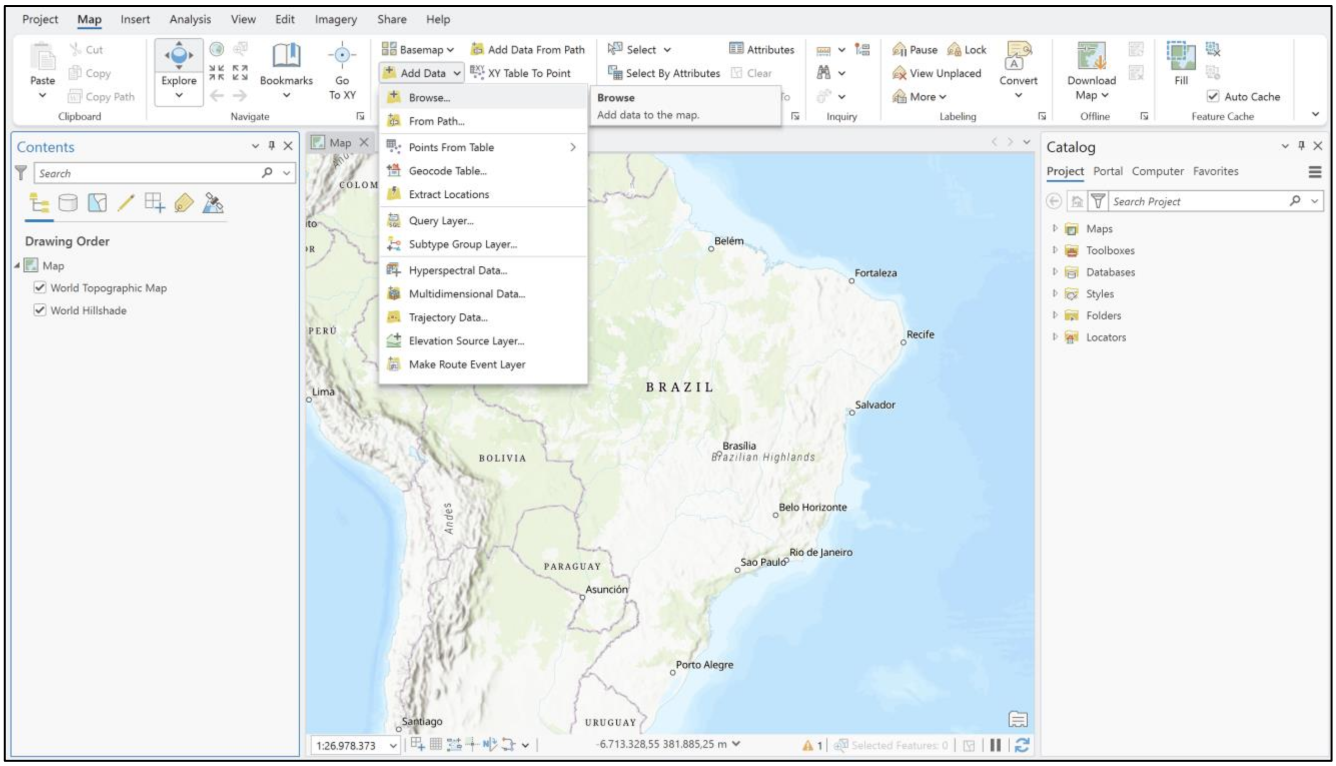
Now your groundwater-well dataset is cleaned and ready to be imported into ArcGIS!

5. Loading and Processing Data in ArcGIS

Launch ArcGIS Pro (3.7.0) and create a new project (New Project). There will appear a blank page or a map already in the background.

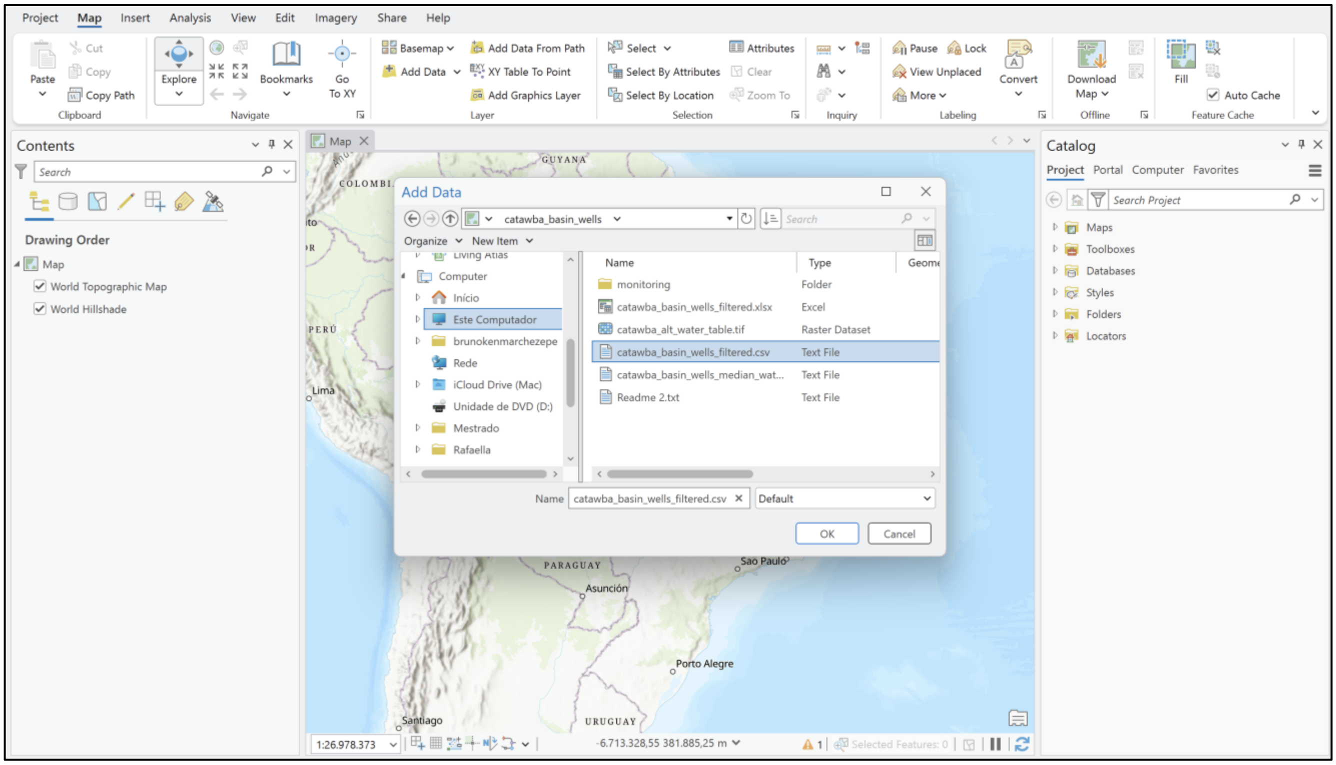
5.1. Adding the vector layers:

1. Navigate to: **Map** → **Add Data** → **Browse**

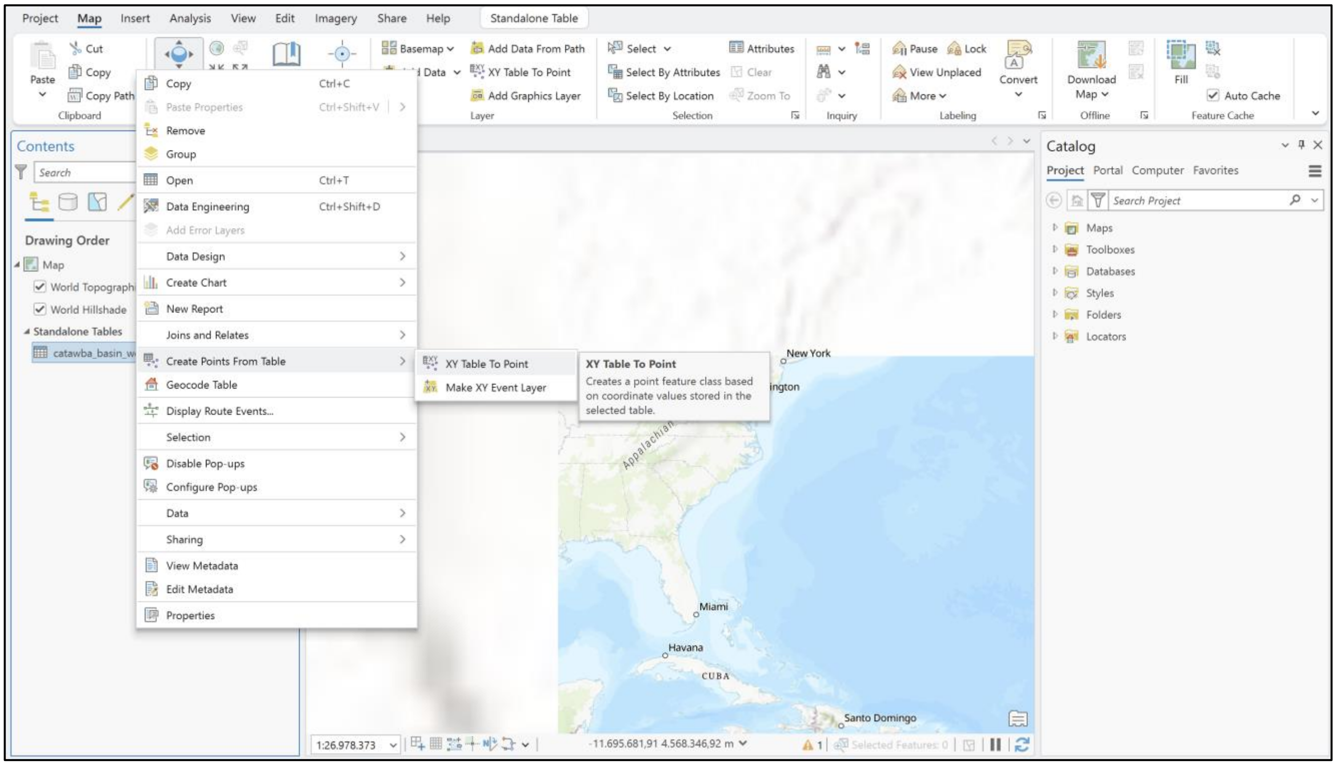


Note: You will see that in **Add Data**, there is an option of **Points from Table**, that we will use in the next steps. You might use this option, but here, we are adding with **Brouse** just so you to know that you can also add other data formats.

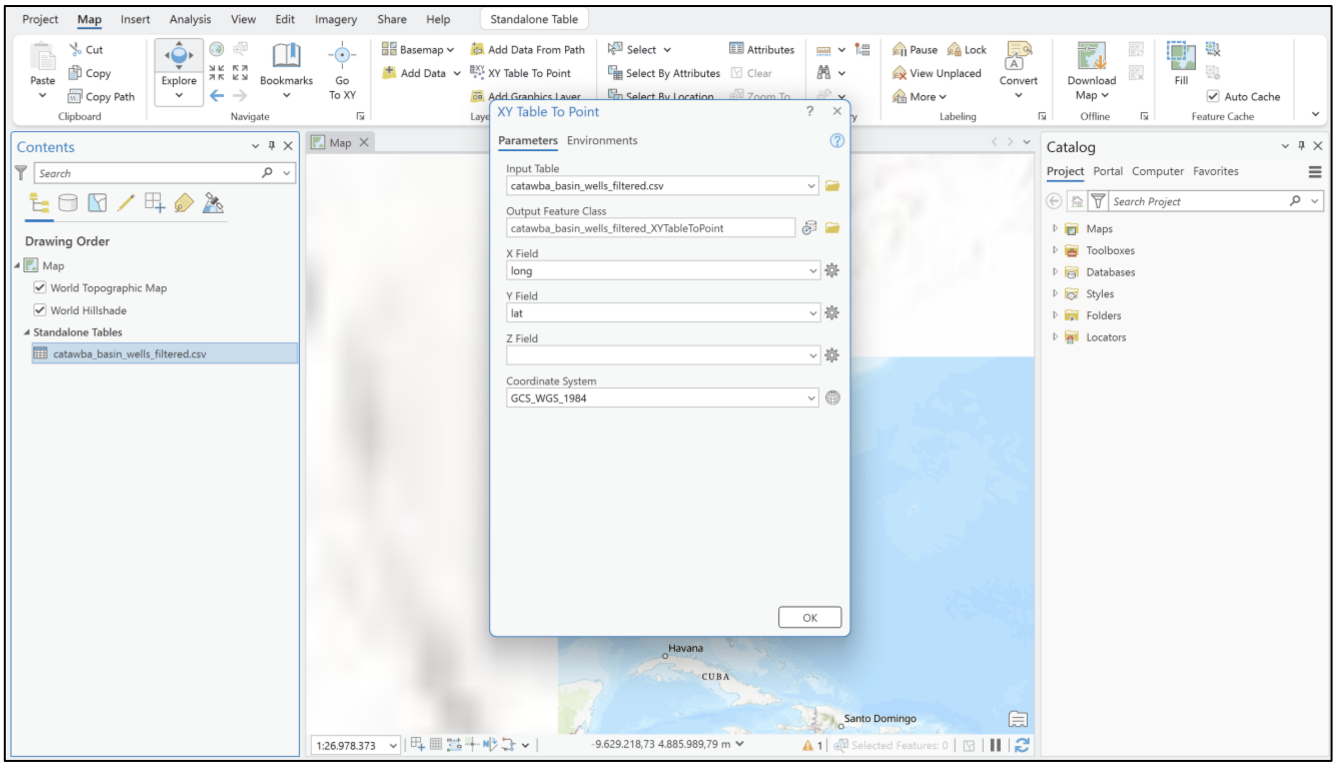
2. Browse to the working directory and select our *.csv* file. Click "Ok".



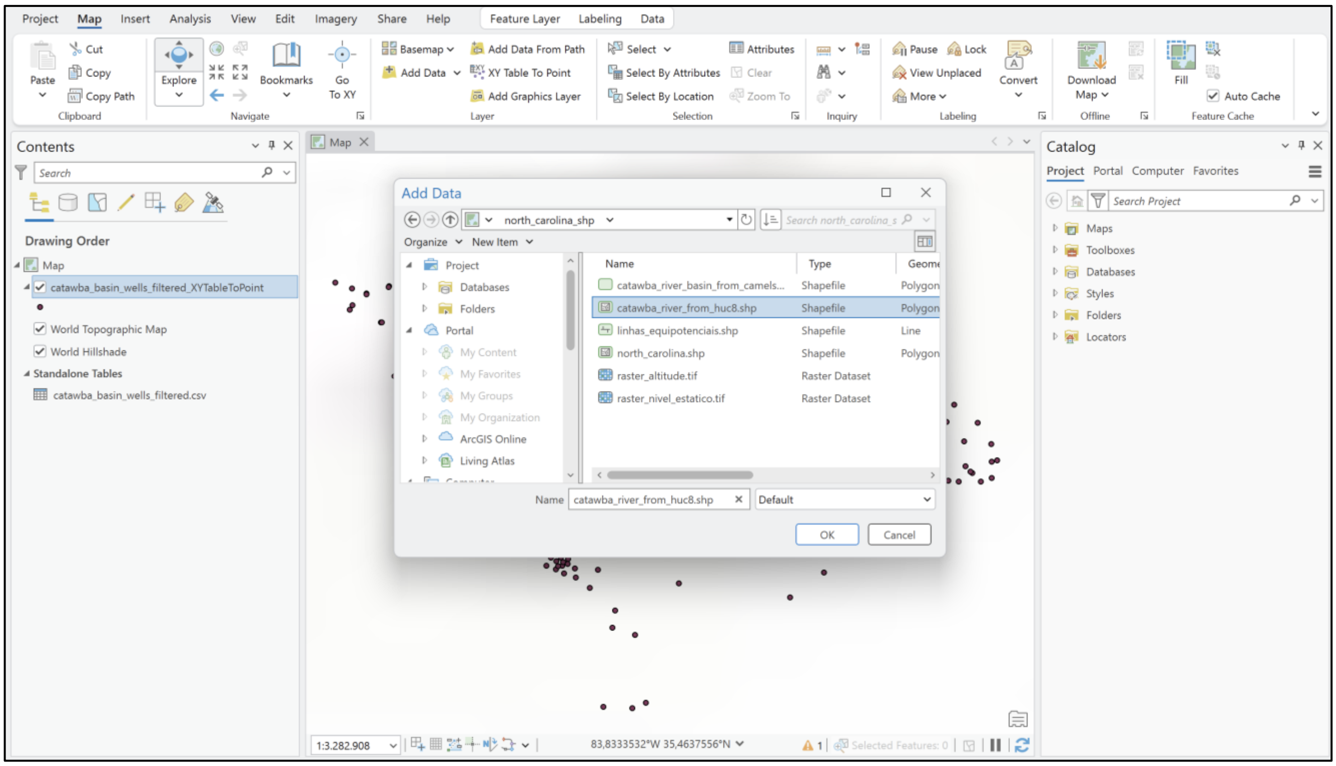
3. The table will appear in the left panel of our project. Right click on the layer and select **Create Points From Table → XY Table To Project**.



4. In this window, you might choose the input table (it is already automatically written), the output name, which might be the ArcGIS suggestion of the ArcGIS or any other name that you want (preferably a name that you will remember). ArcGIS generally suggests the X and Y fields for the points coordinates, but you might choose. Z field might be our elevation, but it is not mandatory for now. Finally, select the Coordinate System (WGS84).

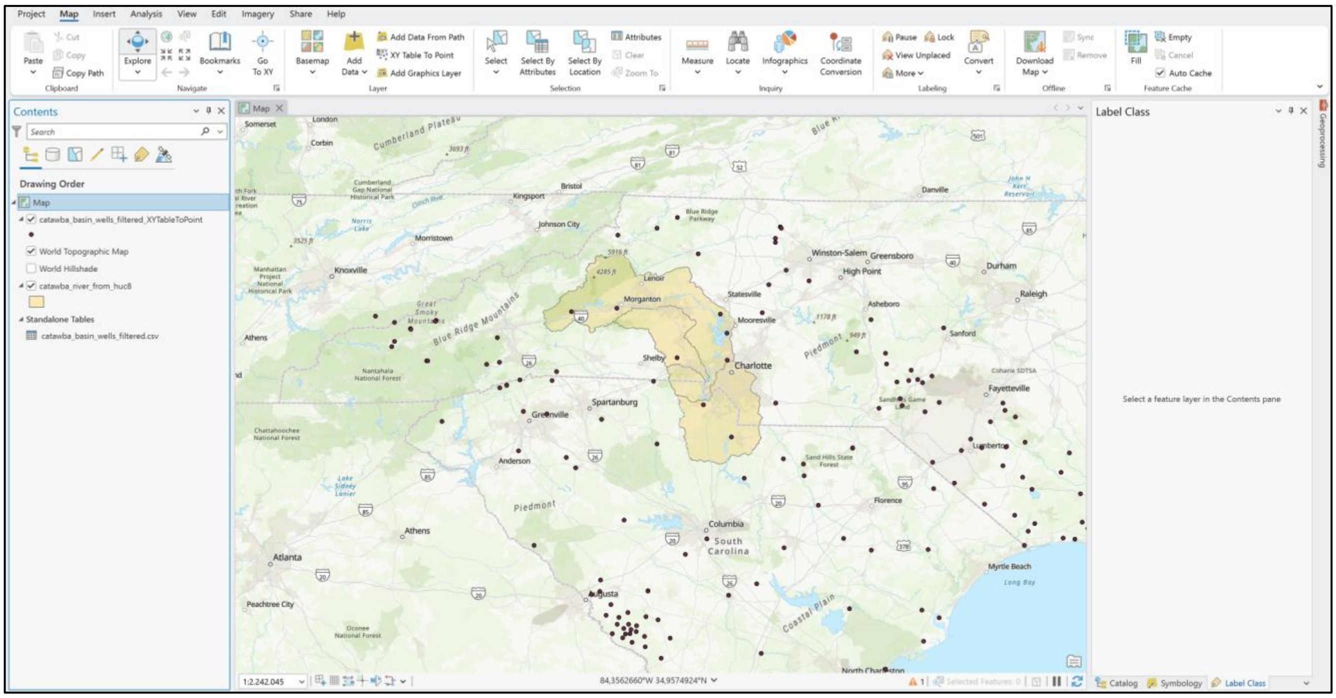


5. Now, go to **Add Data → Browse** and select the Catawba River Basin boundaries in shapefile (.shp). It is available for downloading at https://drive.google.com/drive/folders/1oS9evLg2coisVHVaKsoHfgr34eg5MQCx?usp=sharing.



Note: If you want to analyze other HUC8 basins, please feel free to download the boundaries here: https://www.hydroshare.org/resource/b832a6c2f96541808444ec9562c5247e/

6. Now we already have the wells and the boundaries needed for our analysis. Note that we got more points outside the boundaries than inside them. There is no problem with that, because we will generate the interpolations and then cut the images with the basin boundaries.



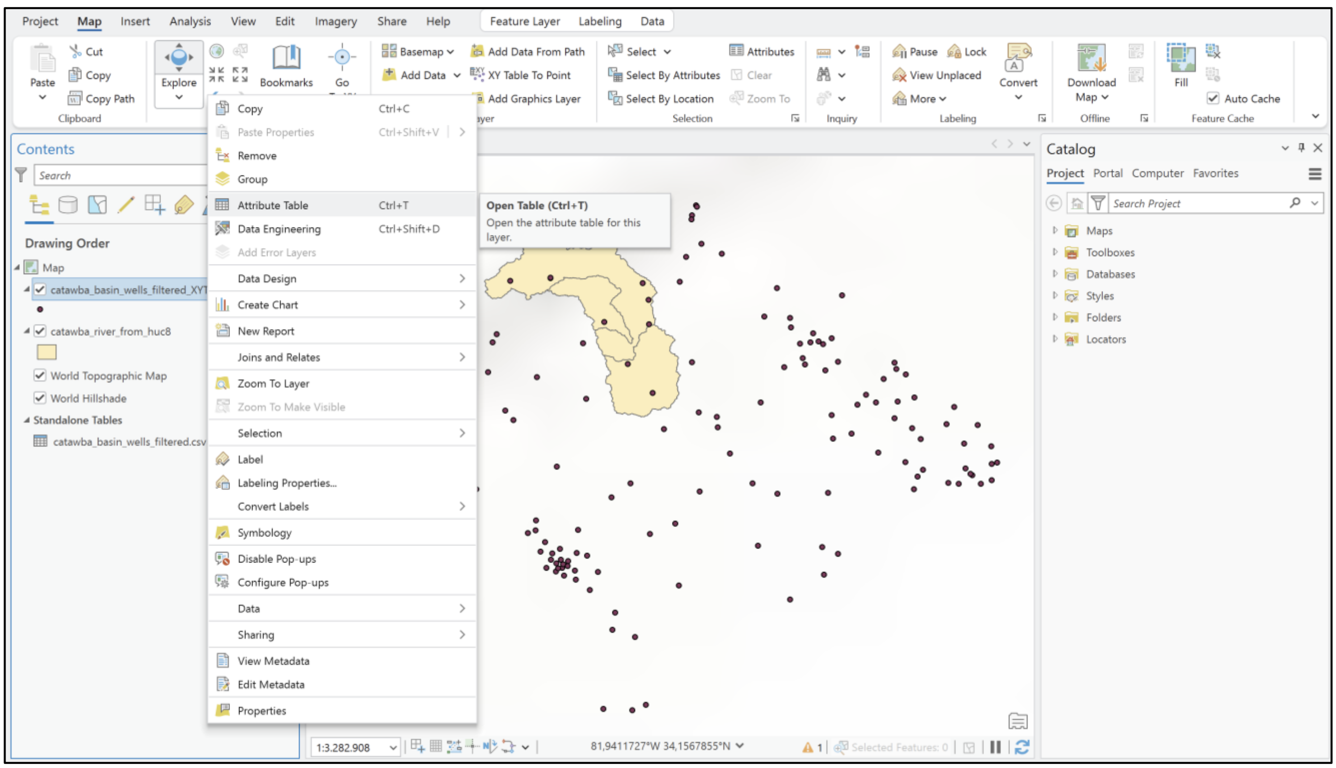
Important: Ensure all working vector layers are projected/set to WGS84 (EPSG: 4326). Otherwise, you must reproject it to not have any trouble while importing or handling the layers.

5.2. Calculating the Water Table / Potentiometric Head

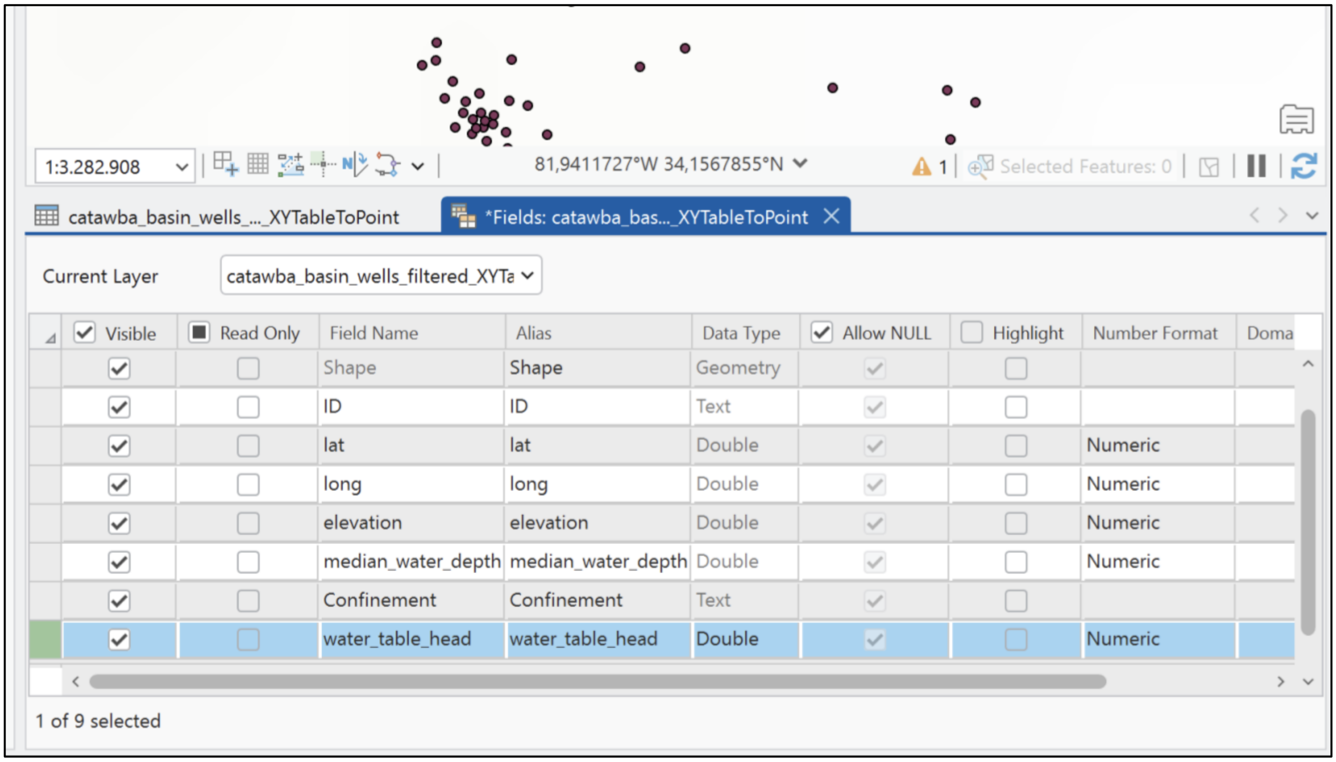
Groundwater hydraulic head / water table elevation is calculated by subtracting the static water level depth (Static Water Level – SWL) from the surface ground elevation at each well point:

> **Head Calculation Formula:** Water Table Head (WTH) = Ground Surface Elevation (GSE) − Static Water Level Depth (SWL)

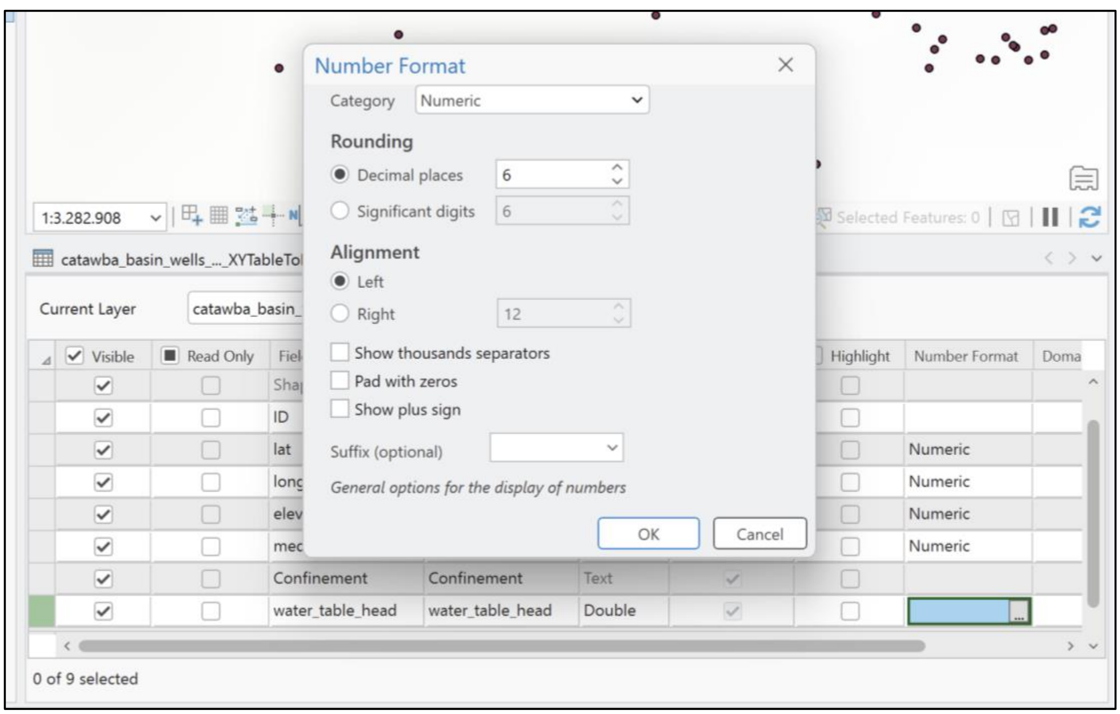
1. Right-click on catawba_basin_wells_filtered_XYTabletoPoints and select **Attribute Table**.



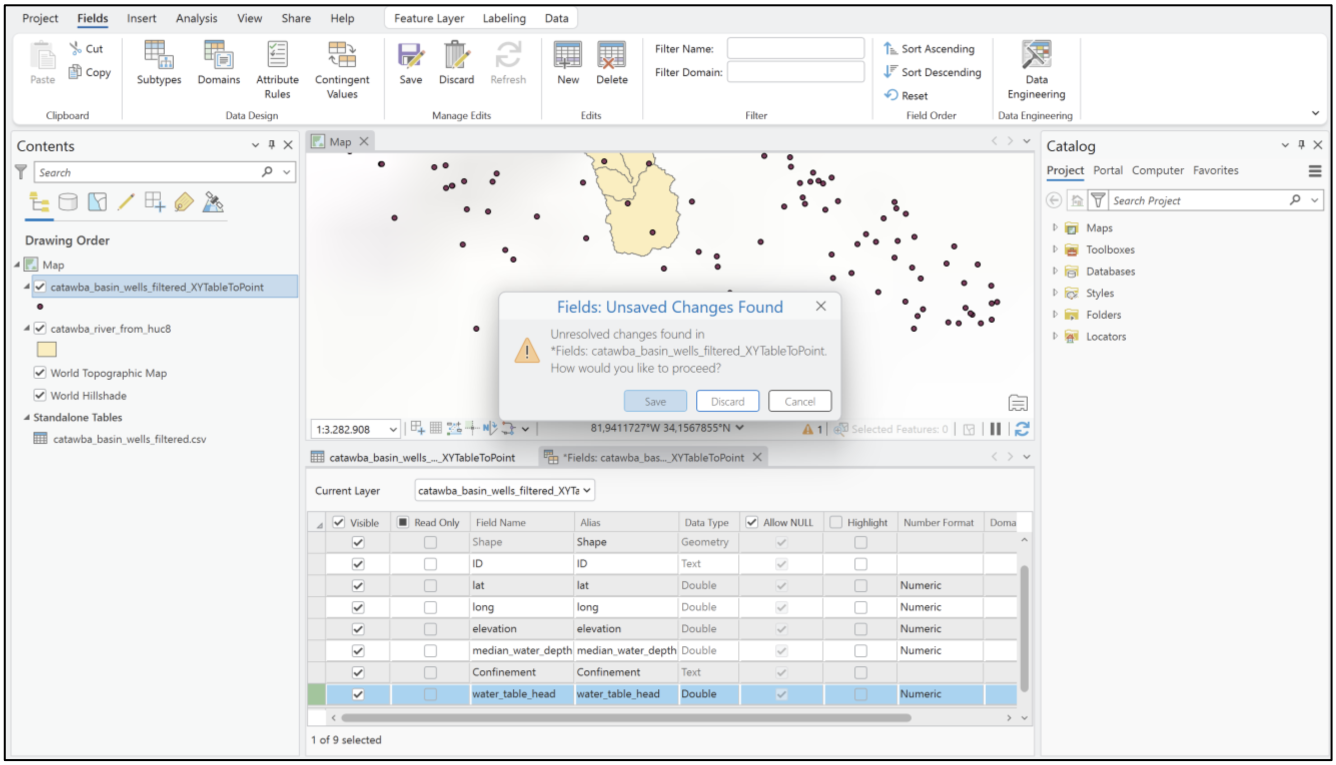
2. Notice the table with one line per well. Click on **Add** to add another column. A new blank line will appear in the metadata table. You can complete as written below.



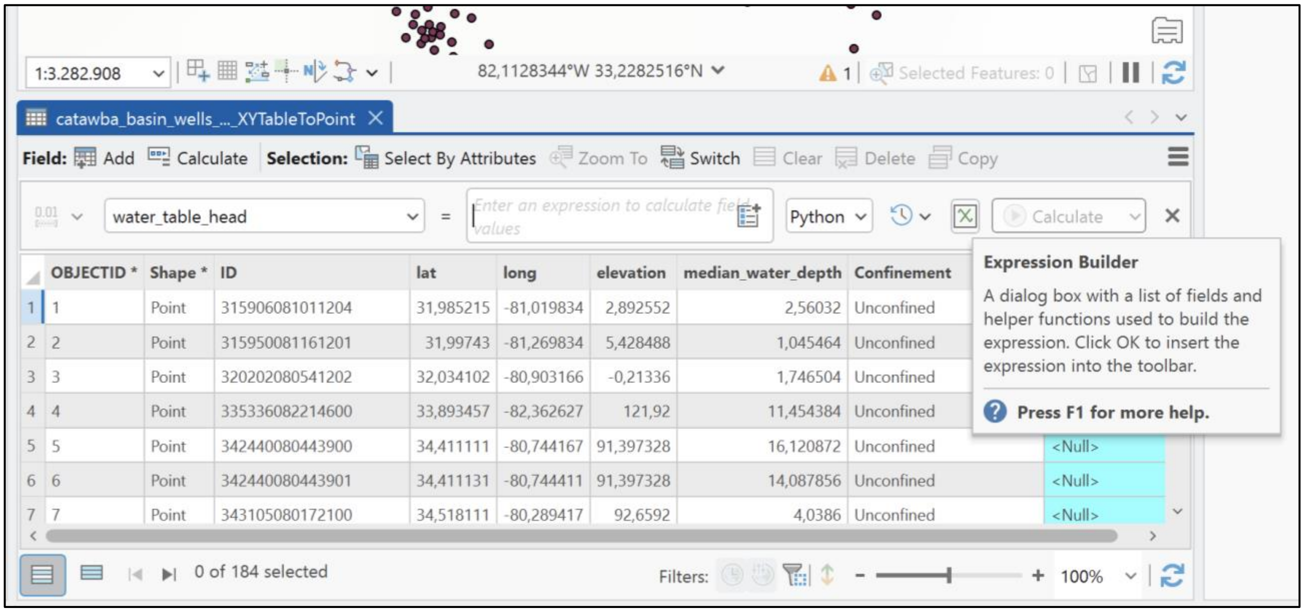
Note: on **Number Format**, you can fill the data as Numeric, with the default format in the following window.



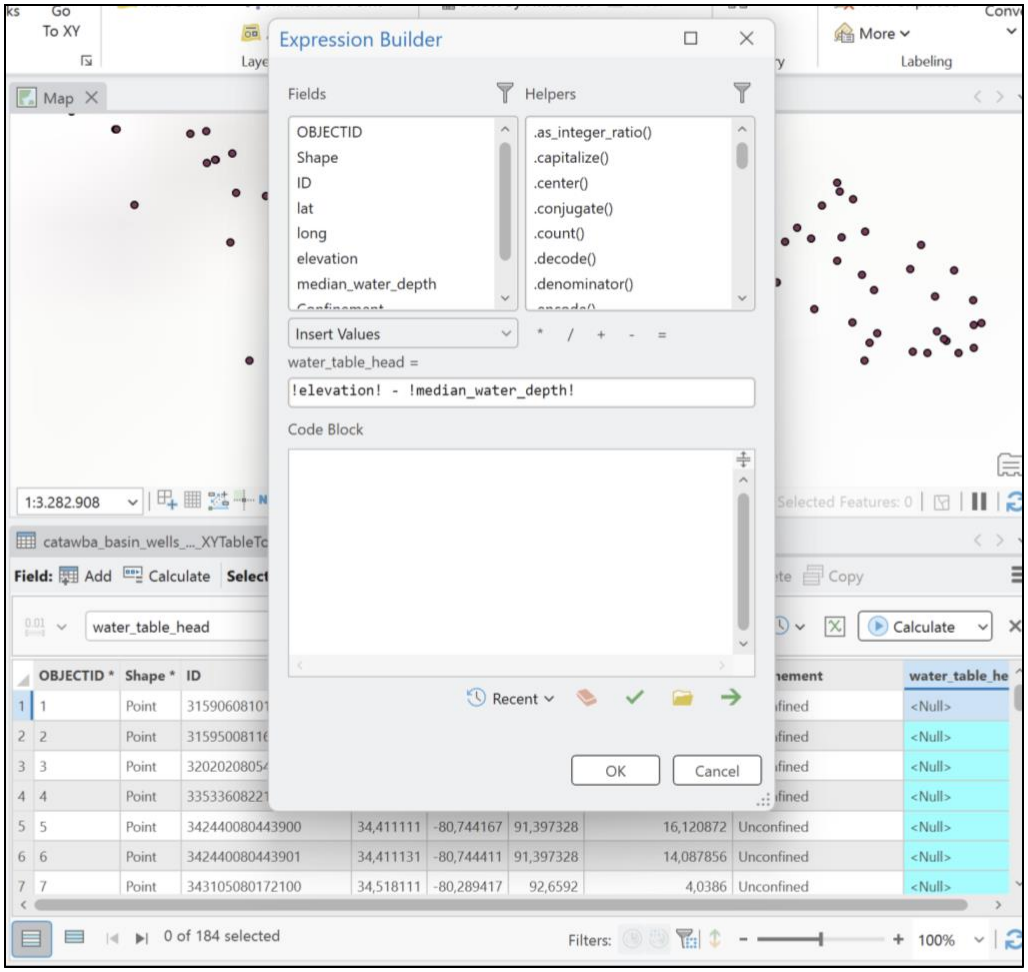
3. Close the metadata table and save it.



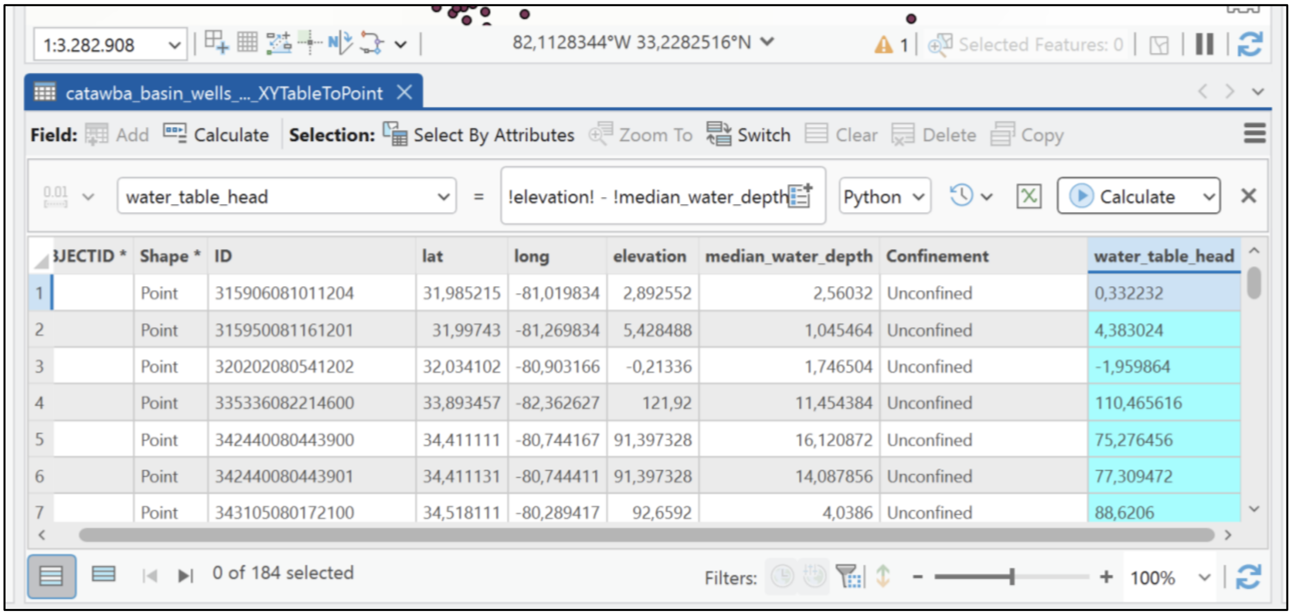
4. The new column will appear in the table. Click the **Calculate** buttom.
5. The calculator will appear right in the top of the table. Choose the water_table_head and click the buttom of **Expression Builder**.



6. For this window, select the same equation as follows: **"!elevation!" - "!median_water_depth!"**.



7. Click **OK**. The newly computed head values appear in the water_table_head attribute column.

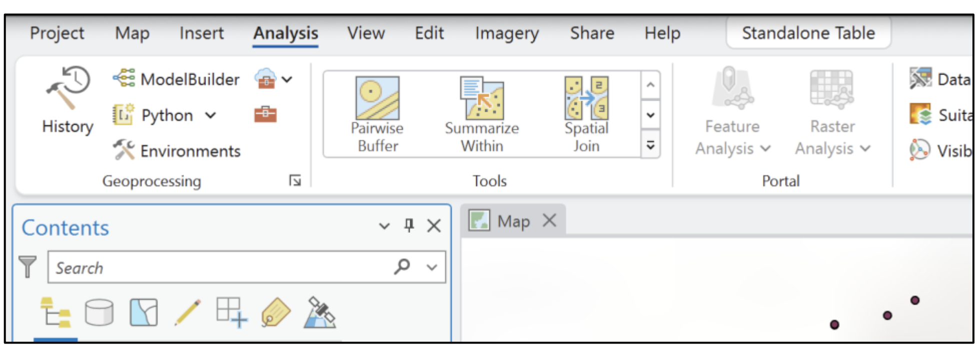


5.3. Spatial Interpolation and Generating Equipotential Lines

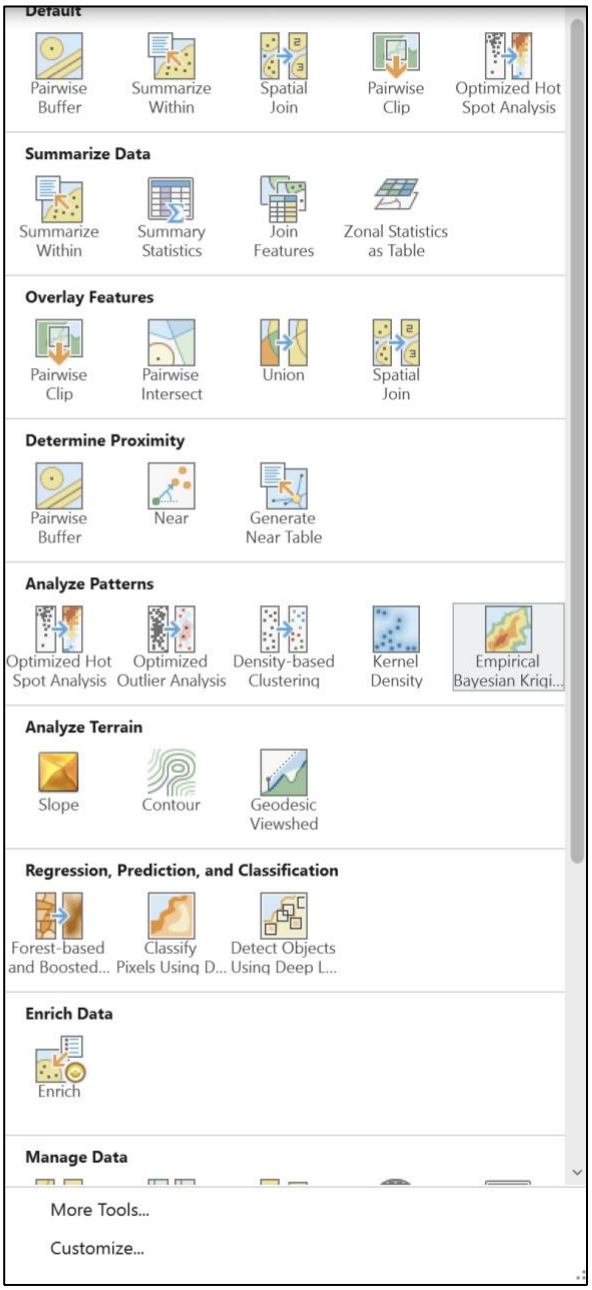
Now, we will analyze 2 methods to interpolate the water table head: Thin Plate Spline (TPS) and Inverse Distance Weighting (IDW). It is worth mentioning that IDW weights points based on proximity, creating sharp local peaks, while TPS fits a smooth bending membrane-like surface through points, minimizing surface curvature. That is why TPS is a preferred method, generally.

5.3.1. Option A: Thin Plate Spline (TPS) — Preferred Method

1. Open the **Analysis** tab
2. Navigate through the **Tools**



3. Go to **Empirical Bayesian Kriging** and fill in the form for **Geoprocessing**. The input feature will be our points, Z value is the water_table_head, and we may choose the output raster name. We call it TPS_interpolation. The output cell size might be as small as we want (which may help in better resolution), but as cells become smaller, the time for computing increases. Choose the semivariogram model type as **Thin plate spline** and **Run**.



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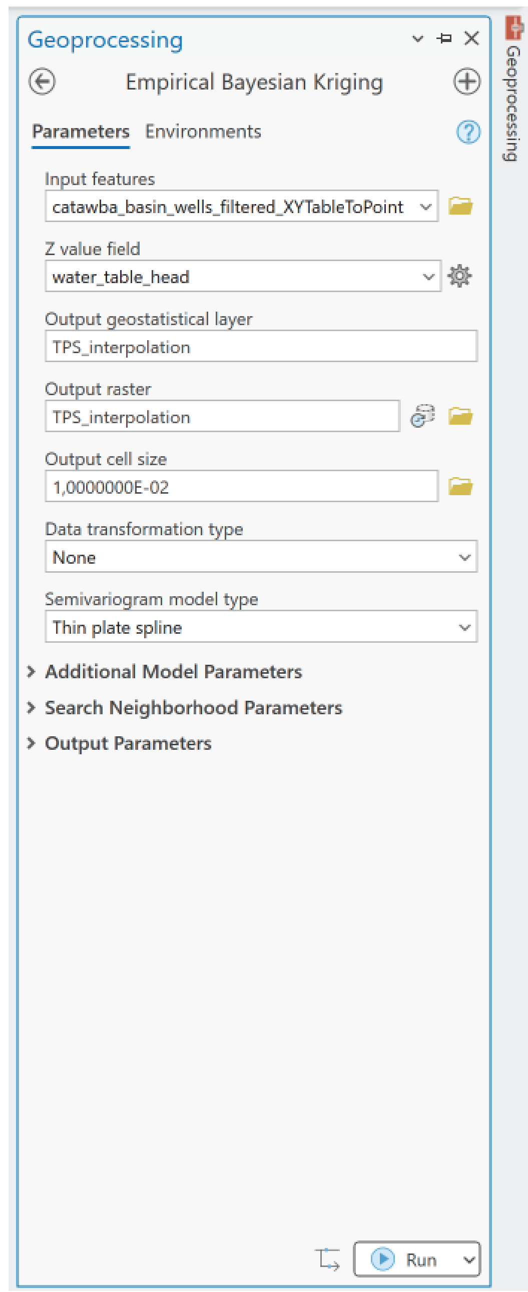
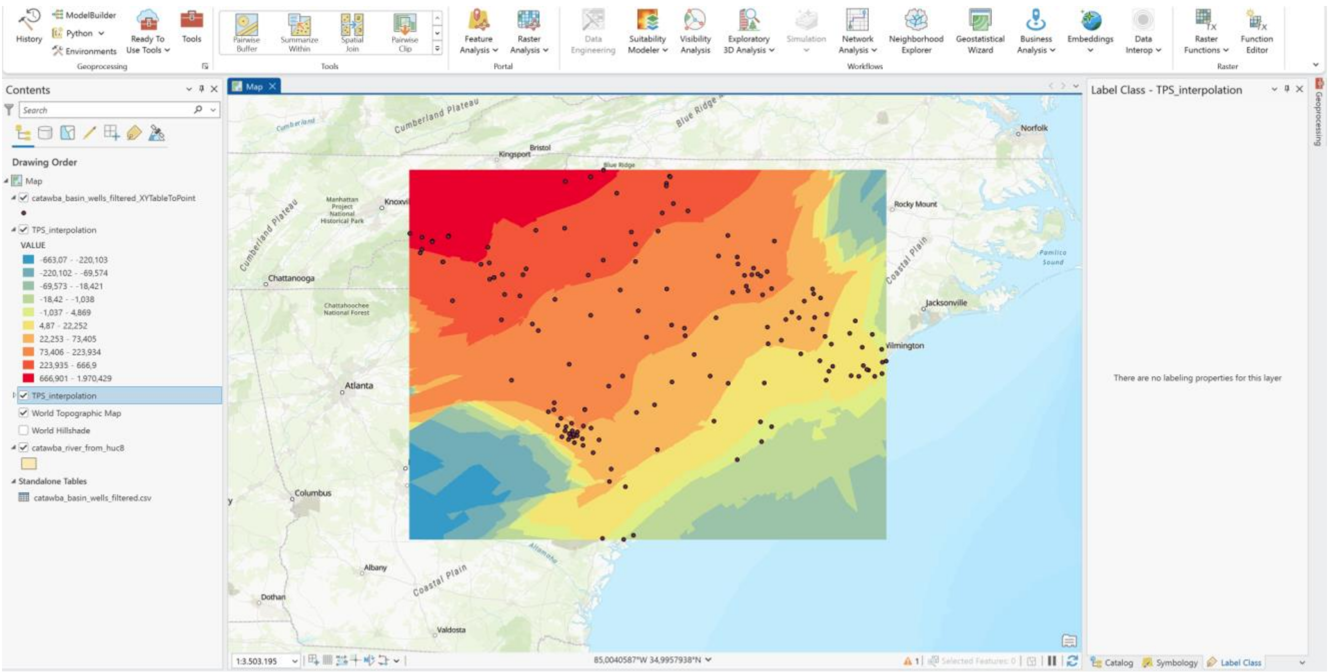
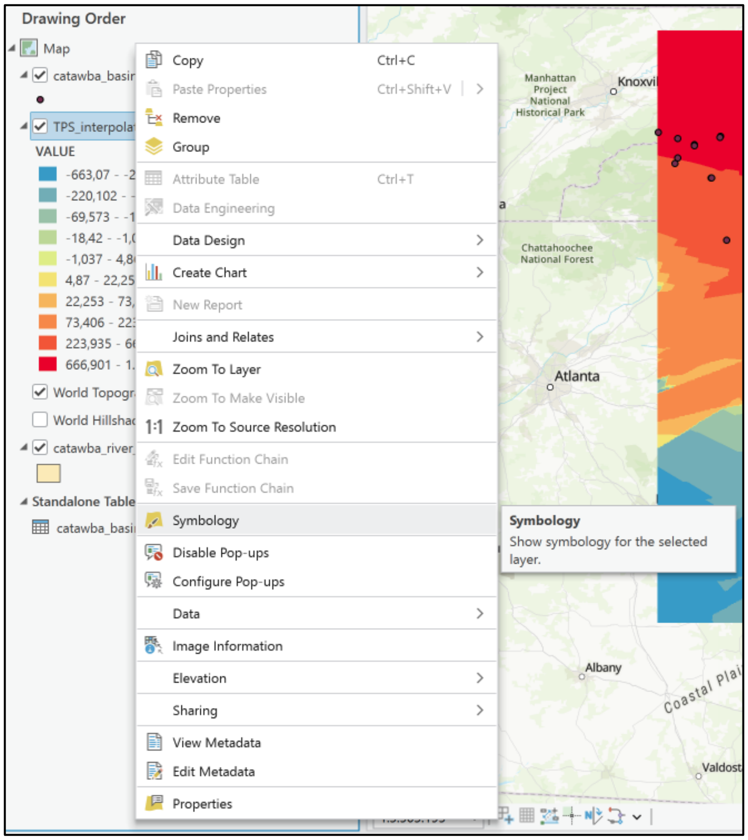


4. The interpolated raster is shown below. Note that it extrapolates the values where there are no wells.



5. Now we can change the **Simbology** of this raster. Right click on the raster layer and choose the values of classes. We can change the number of classes, the method to calculate the intervals and choose the upper values and label for each class.



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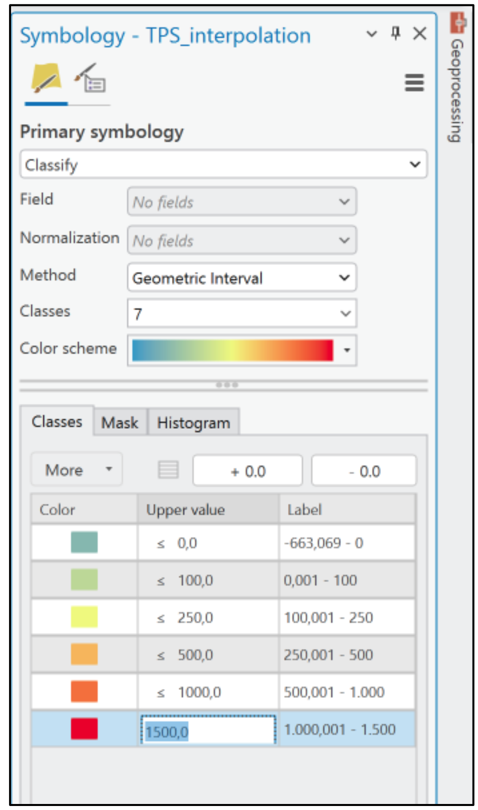
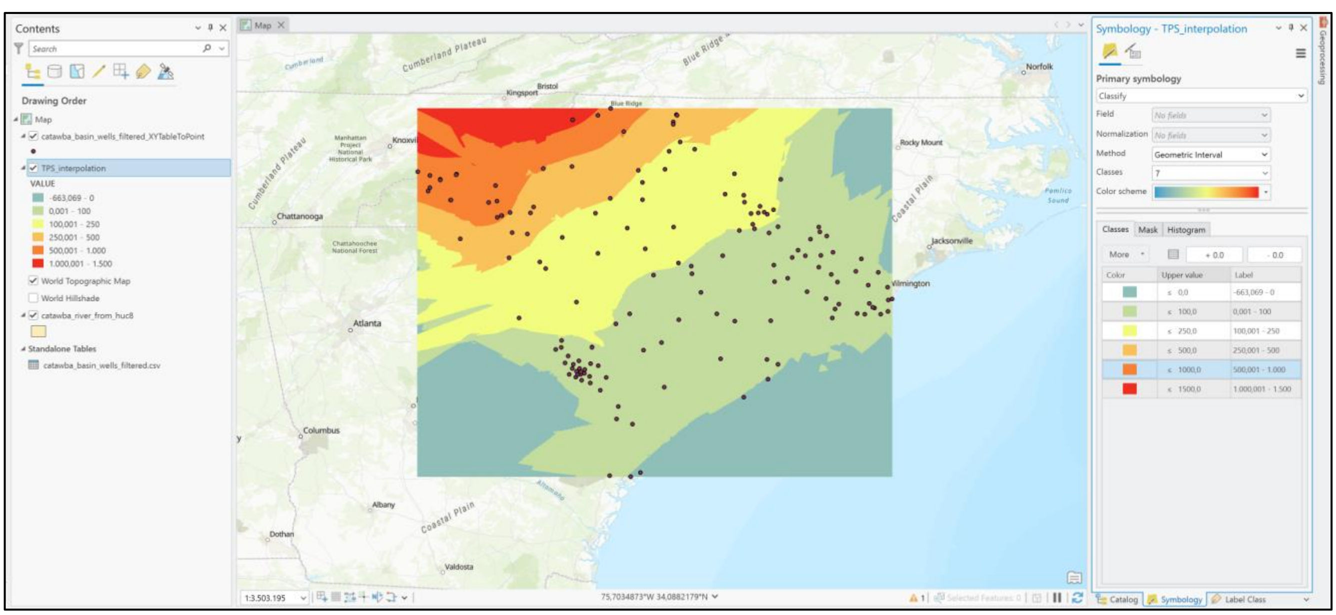
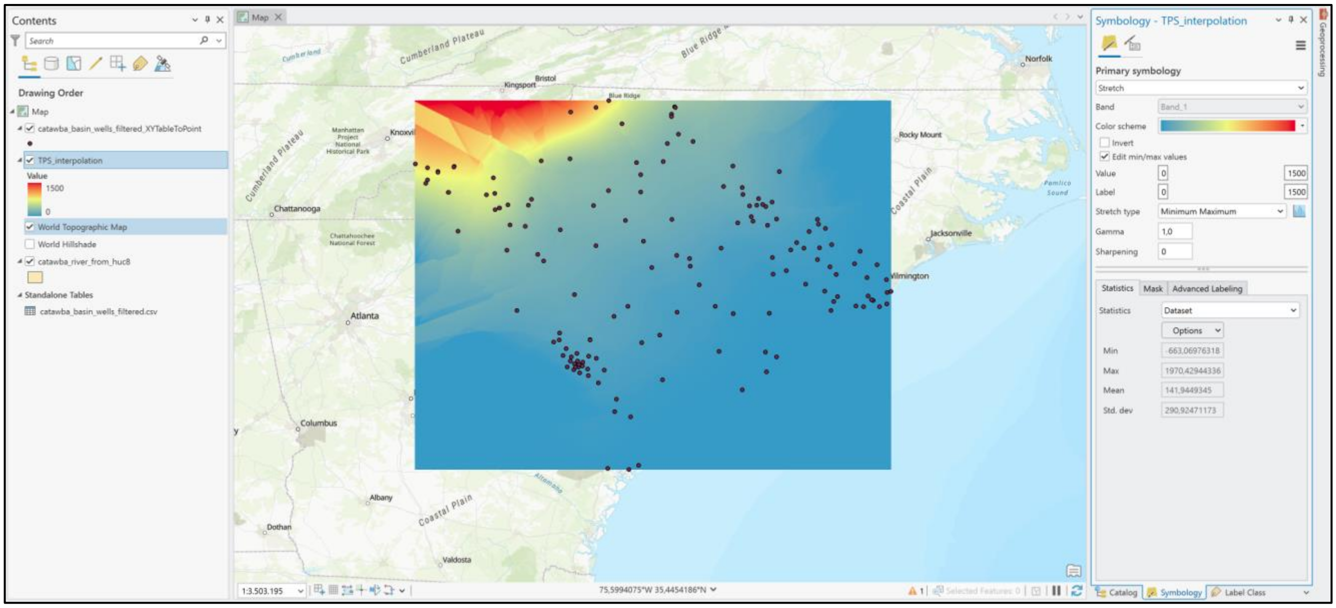


6. The results for classification interpolation is shown below, according to our intervals.

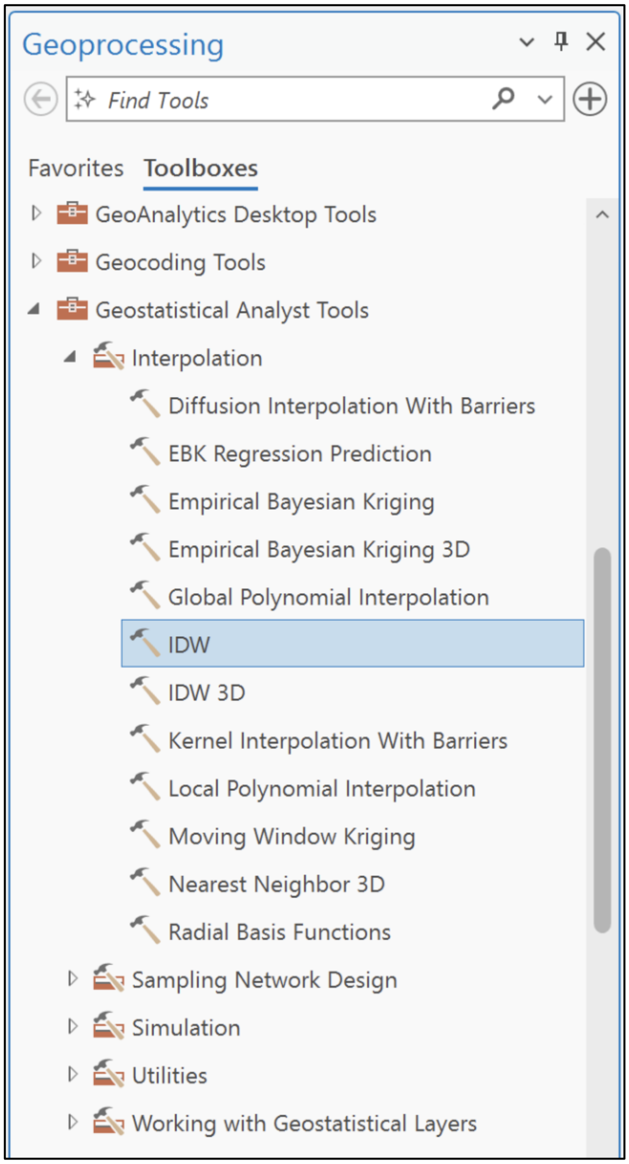


7. We can also change to a stretched band of colors.

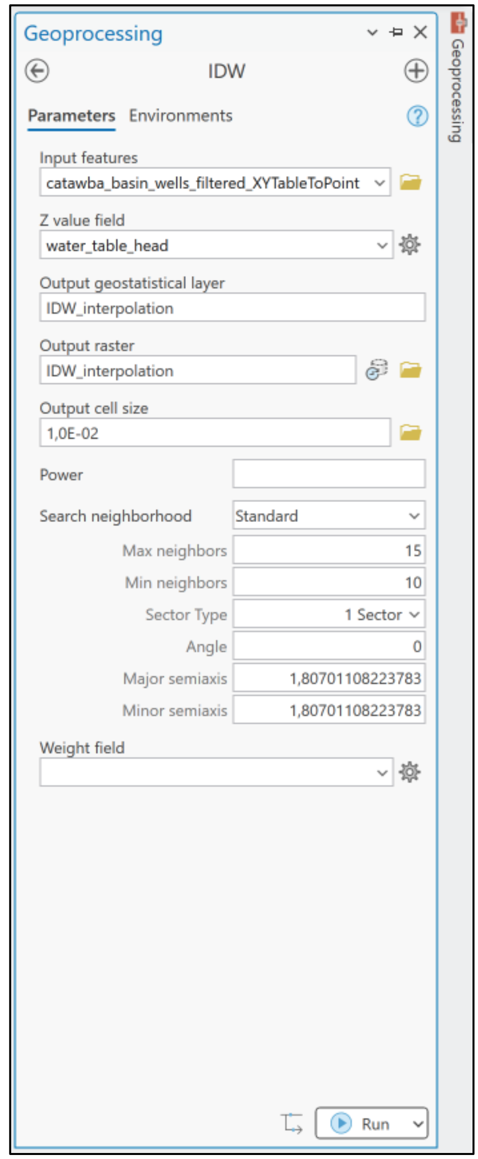


5.3.2. Option B: Inverse Distance Weighting (IDW)

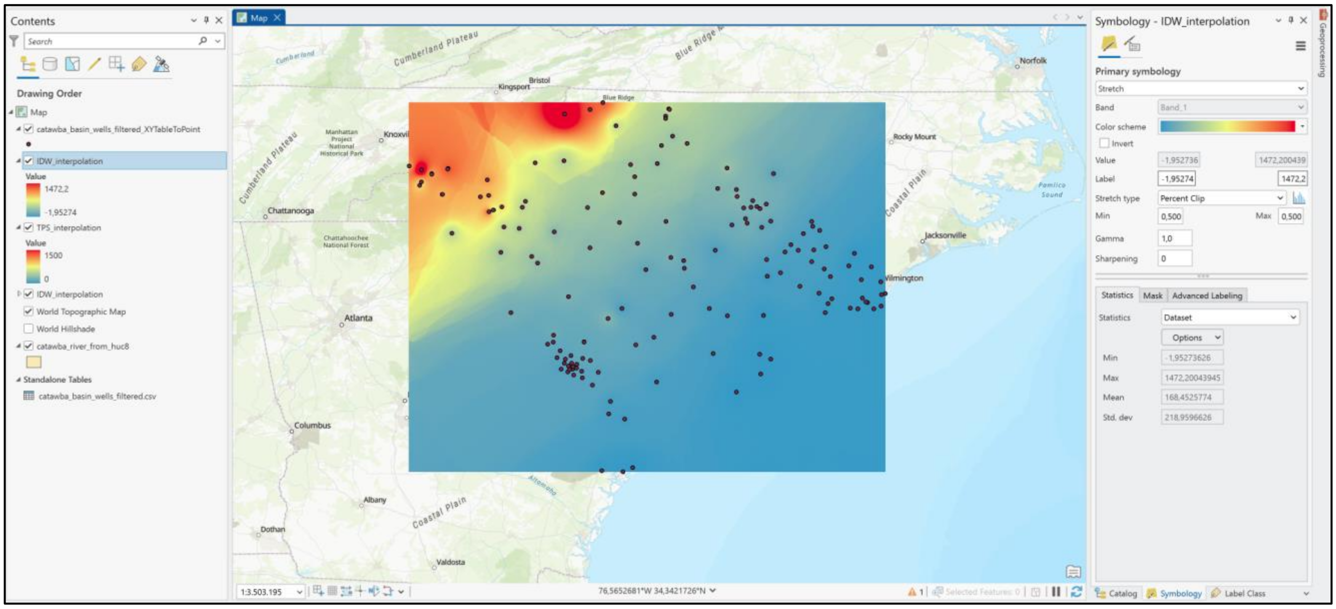
1. Almost the same as TPS, go to **Analysis → Tools** but now, scroll to the end and click on **More Tools…** This opens the **Geoprocessing** tools. Choose **IDW**. Fill the box with the points layer, water_table_head for Z value field, the name IDW_interpolation, cell size of 1E-02 and let the others by default (for further analysis, you might see the functionalities by passing the cursor in the little "i" next to the name of each field.


Geoprocessing
Find Tools
Favorites
Toolboxes
GeoAnalytics Desktop Tools
Geocoding Tools
Geostatistical Analyst Tools
Interpolation
Diffusion Interpolation With Barriers
EBK Regression Prediction
Empirical Bayesian Kriging
Empirical Bayesian Kriging 3D
Global Polynomial Interpolation
IDW
IDW 3D
Kernel Interpolation With Barriers
Local Polynomial Interpolation
Moving Window Kriging
Nearest Neighbor 3D
Radial Basis Functions
Sampling Network Design
Simulation
Utilities
Working with Geostatistical Layers

→

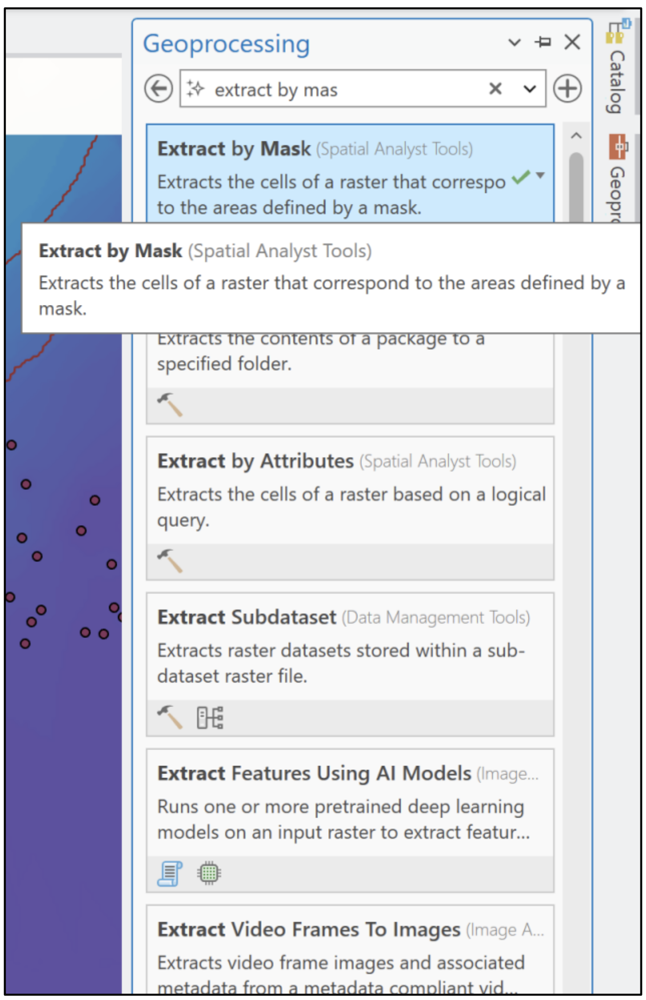

Geoprocessing
IDW
Parameters
Environments
Input features
catawba_basin_wells_filtered_XYTableToPoint
Z value field
water_table_head
Output geostatistical layer
IDW_interpolation
Output raster
IDW_interpolation
Output cell size
1,0E-02
Power
Search neighborhood
Standard
Max neighbors
15
Min neighbors
10
Sector Type
1 Sector
Angle
0
Major semiaxis
1,80701108223783
Minor semiaxis
1,80701108223783
Weight field
Run
Geoprocessing

2. The result will be similar to the one from TPS, but you might see exactly what we discussed about the IDW method that might create local peaks that are not so coherent with the general view that we need.



5.3.3. Clipping Contours to the Aquifer Domain:

1. We will analyze the TPS_interpolation, as it shows smoother interpolation surface
2. When in **Geoprocessing** tools (**Analysis → Tools → More Tools…**), search for Extract by Mask. Fill the input raster (TPS_interpolation), the mask layer (Catawba River Basin boundaries), the output name and if the extracted area will be inside or outside the mask. **Run**.



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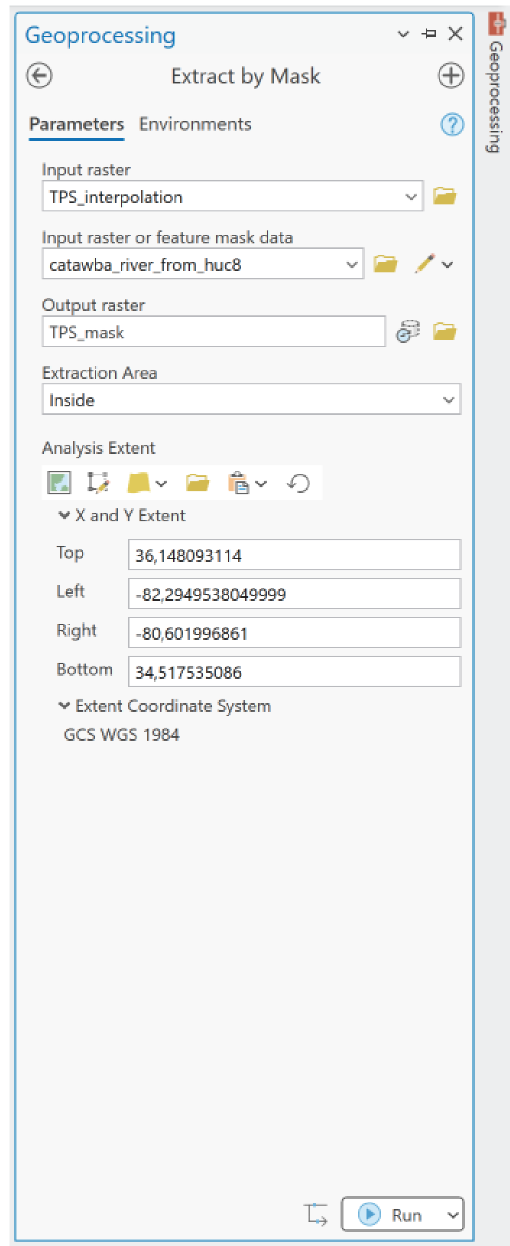
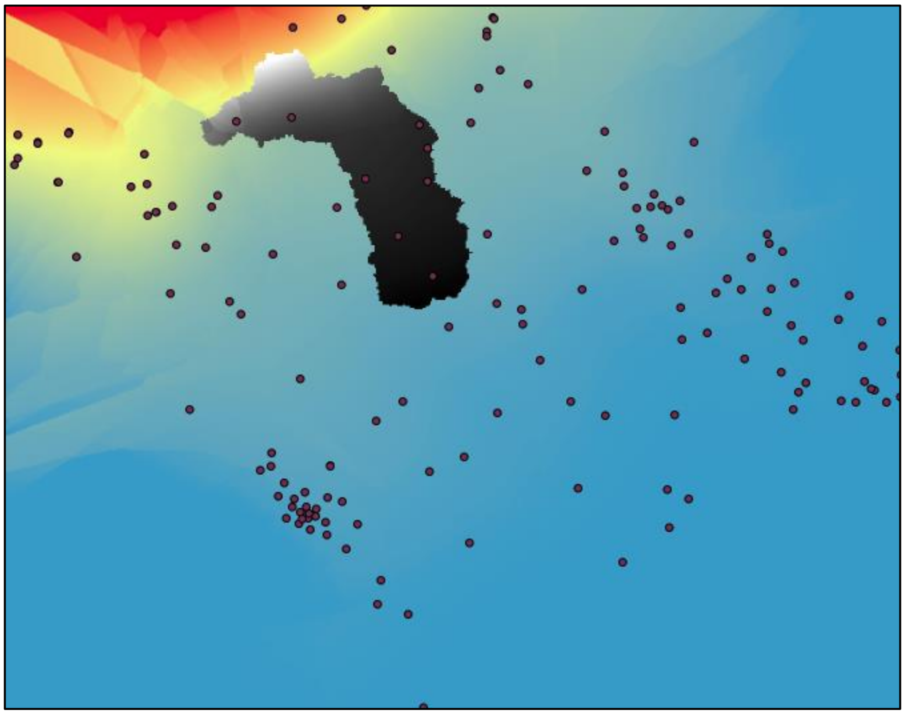
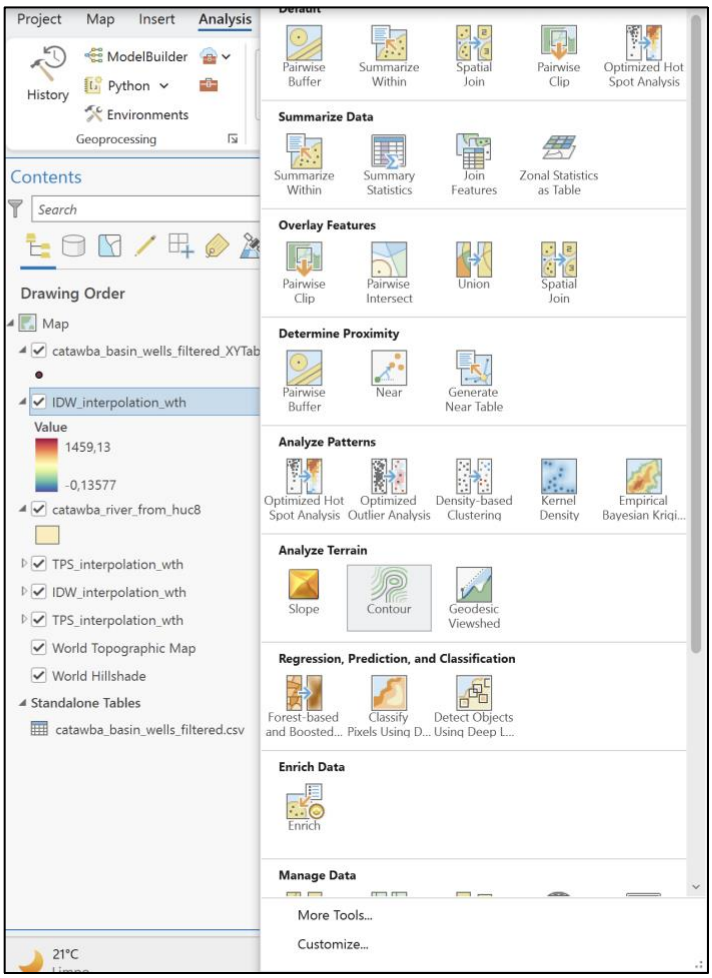


3. Now we have a new raster with the basin contours. You can chance the **Symbology** as you want. Also, you might want to hide some dispensable layers, such as the original interpolation raster.

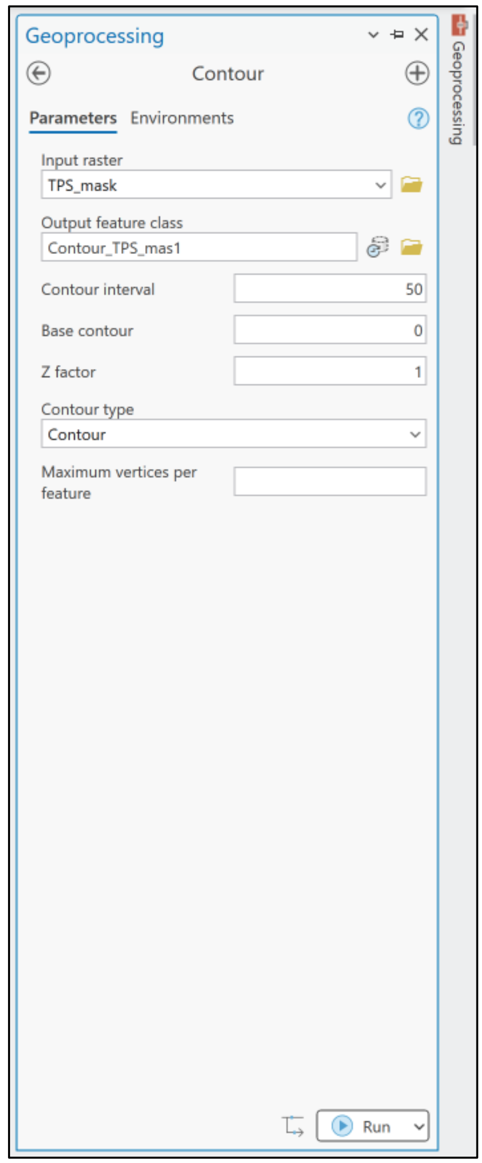


5.3.4. Extracting Equipotential Contours

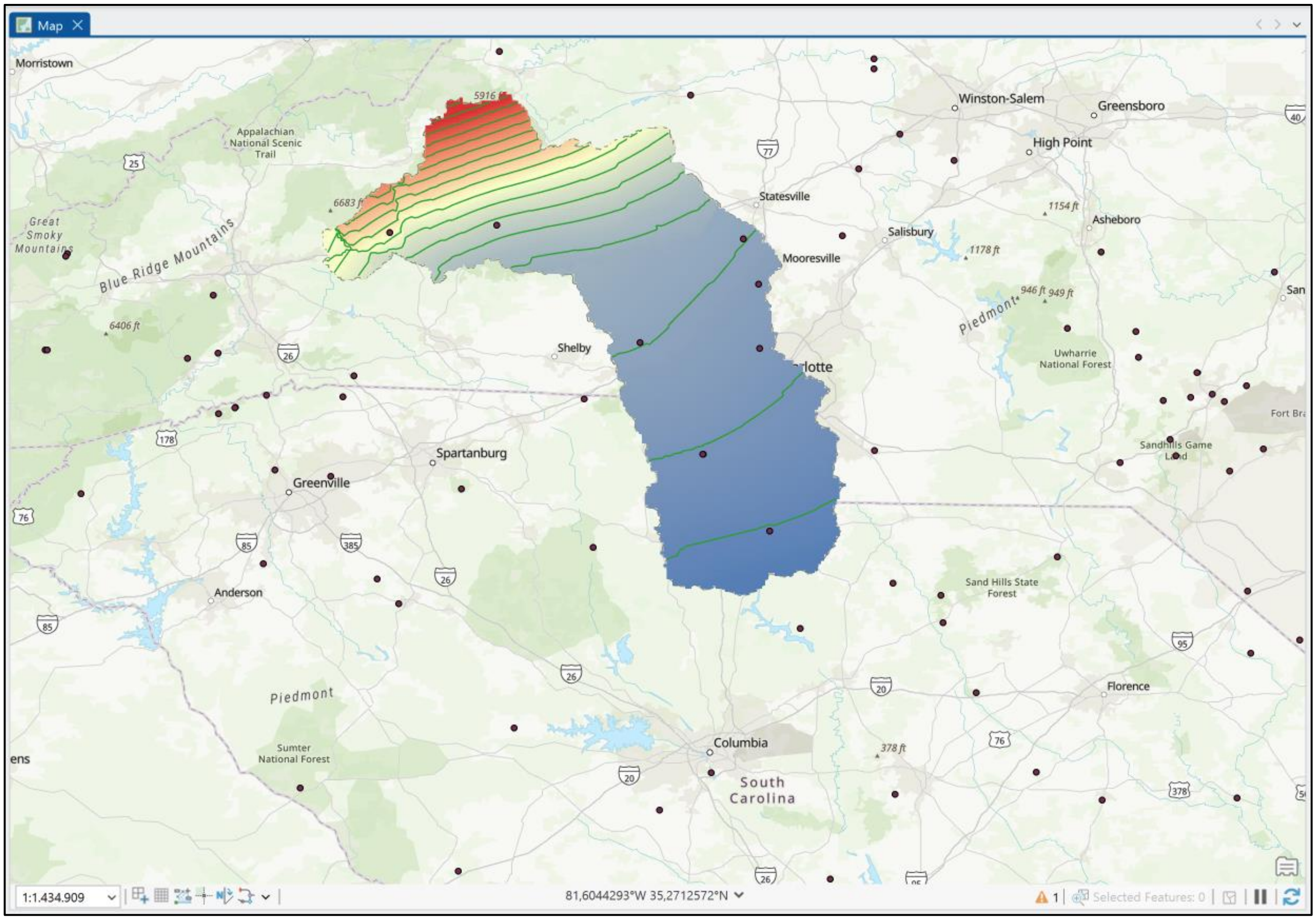
1. At last, we will insert the equipotential contours. Go to **Analysis → Tools → Contours**. Fill in the input raster (TPS_mask) and the Contour interval, which represents the vertical distance between two consecutive contours. In this case, you can put 50 meters. **Run**.



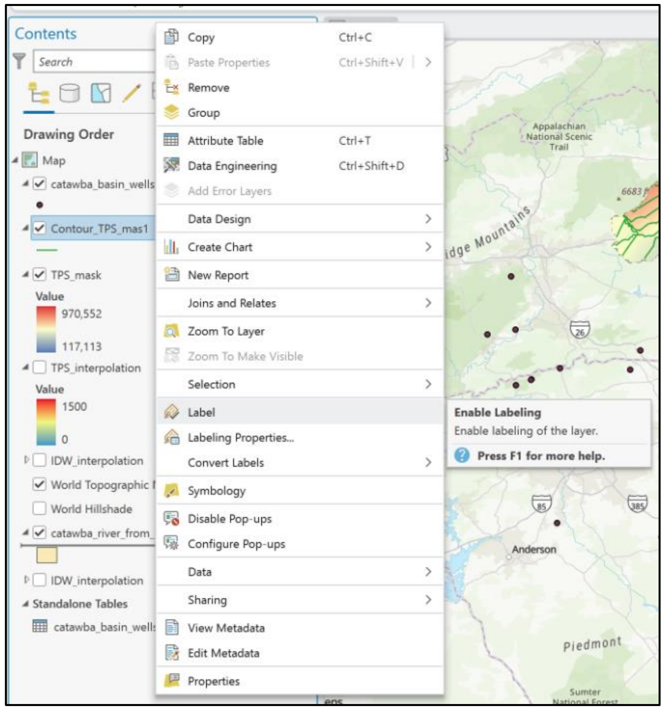
→



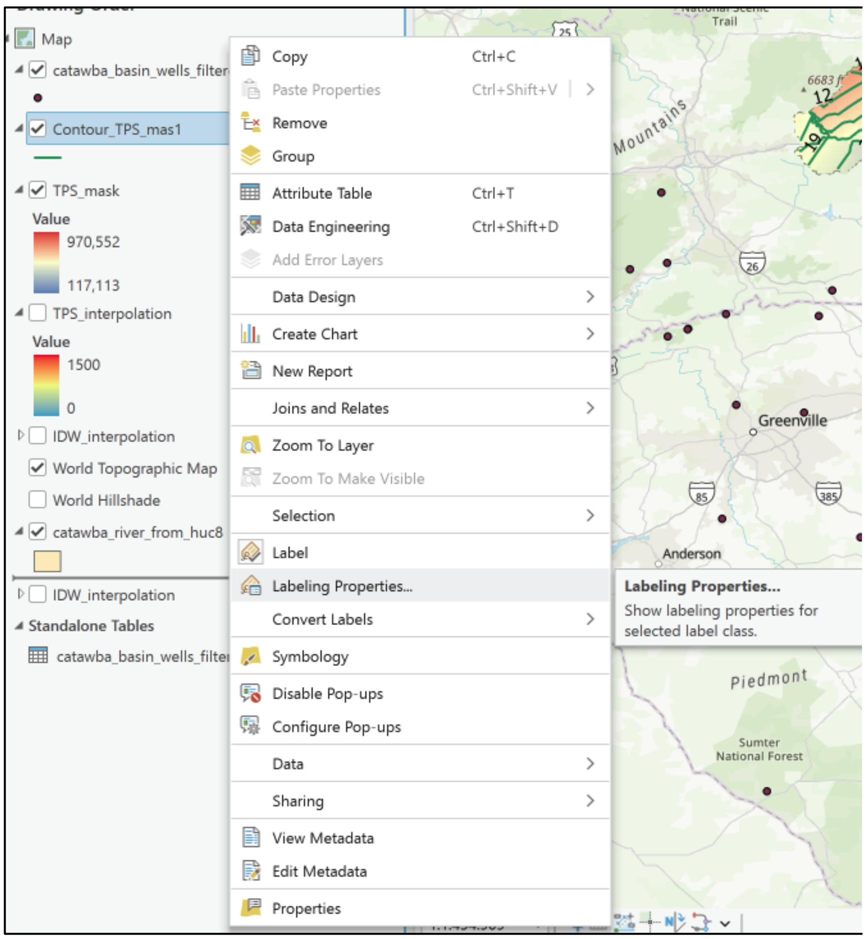
2. Almost done, you should get something like the next figure.

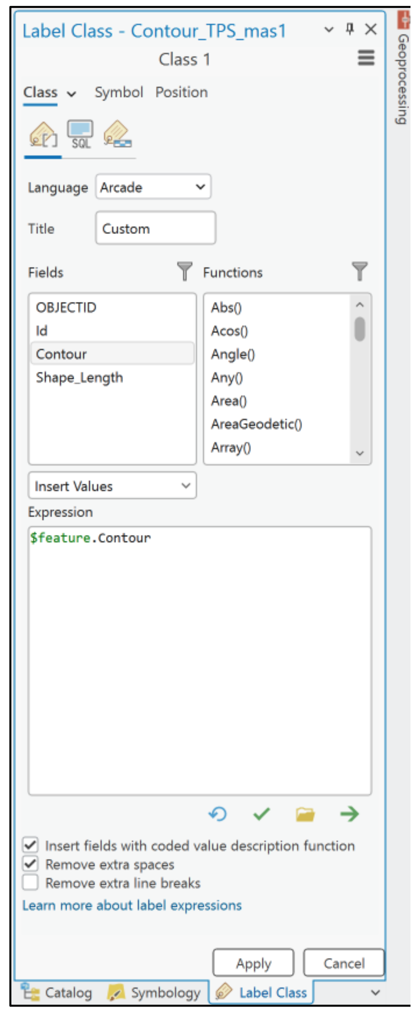


3. Let's put labels to the contours. Right click on the contour layer, and **Label**.

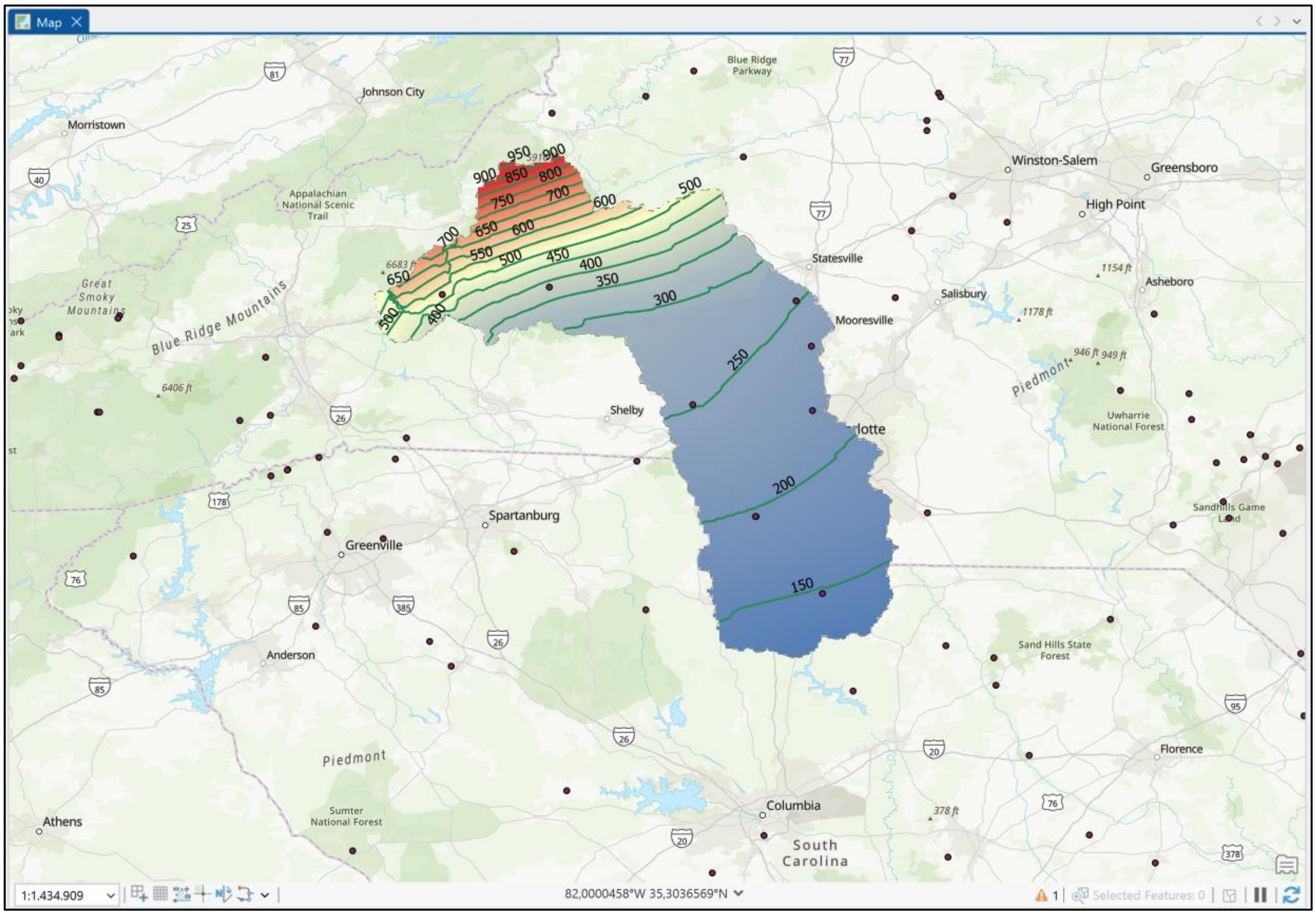


4. You will get the labels, but only for the ID of each contour. We want the Water Table Head. So again, Right Click on the contour layer, and **Labeling Properties…** When filling the fields of properties, substitute the **$feature.Id** for **$feature.Contour**.





5. Your potentiometric map is ready! If you want, you now can make a print layout now. Check the https://youtu.be/tmi1V2xk2JY?si=p3qUvoczFmjutXbI for some quick ideas.



6. Summary of Activity Requirements (from Assignment Guidelines)

Report Section	Specification / Content Requirement	Limit
Cover Page	Header (University, School, Course), Student full names in alphabetical order with Student IDs, Assignment Title, Course Name, Location (City/State), Year.	1 page
Theoretical Synthesis	Answer key theoretical questions: • What is a potentiometric surface map? (Max 4 lines) • Why must an Environmental Engineer know how to build and interpret potentiometric maps? Cite 3 real-world engineering project applications. (Max 6 lines) • What are the limitations of a potentiometric map? (Max 5 lines) • What is the static water level? Why is it inadequate to use static level alone without surface elevation for flow mapping? (Max 5 lines) • What is hydraulic head and how was it derived in this activity? • What is a Digital Elevation Model (DEM)? (Max 3 lines) • Clear theoretical explanation with workflow screenshots, schematics, and proper figure captions.	Max 7 pages
Map Interpretation	Hydrogeological interpretation of equipotential contours and groundwater flow directions.	Max 3 pages
References	Complete formal citations for all databases, software, and academic references used.	No page limit

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